

Beyond Point Estimates: Navigating Paradoxes, Scalability, and Robustness in Deep Learning Uncertainty Quantification

The deployment of deep learning in safety-critical domains necessitates a rigorous transition from deterministic predictions to probabilistic frameworks capable of quantifying confidence. However, standard Bayesian approximations often fail to capture true model ignorance, exhibiting paradoxical behaviors such as the "epistemic uncertainty hole," where uncertainty metrics collapse in large models despite data scarcity [96]. This review synthesizes recent advances addressing these theoretical and practical fissures, moving beyond simple aleatoric-epistemic disentanglement toward robust frameworks like Credal Bayesian Deep Learning, which utilizes sets of priors to mitigate misspecification risks [22]. We critically evaluate scalable inference strategies, including subnetwork inference and last-layer linearization, which preserve uncertainty fidelity in foundation models where full posterior inference is intractable [37]. Furthermore, we explore the emerging paradigm of actionable uncertainty, integrating gradient-based attribution to transform passive confidence scores into interpretable signals for decision-making [19]. By contrasting the theoretical elegance of infinite-width limits with empirical failures in calibration, this work identifies critical gaps in current benchmarking and proposes a roadmap for developing uncertainty-aware systems that are not only scalable but also theoretically grounded and robust to distributional shifts [32].

Introduction to Uncertainty in Deep Learning

The deployment of deep learning models in high-stakes domains such as healthcare, autonomous systems, and climate science necessitates a rigorous transition from deterministic point estimates to probabilistic frameworks capable of quantifying confidence. Standard deep neural networks, while achieving remarkable accuracy, are prone to producing overconfident predictions even when faced with out-of-distribution inputs or corrupted data, a failure mode that can lead to catastrophic consequences in safety-critical applications [8; 173; 57]. This inherent overconfidence underscores the critical necessity of Uncertainty Quantification (UQ), which aims to provide reliable estimates of a model's predictive confidence alongside its output. The foundational challenge in this domain lies in distinguishing between the two primary sources of uncertainty: aleatoric uncertainty, which arises from inherent noise in the data generation process and is irreducible, and epistemic uncertainty, which stems from model ignorance due to limited training data or suboptimal architecture and is theoretically reducible [8; 24; 86]. Disentangling these sources is essential for trustworthy AI, as it allows practitioners to determine whether performance limitations are due to noisy sensors or a lack of model knowledge, thereby informing decisions on whether to collect more data or improve sensor fidelity [79; 120].

Traditional frequentist approaches, which rely on single point estimates of network weights, fundamentally fail to capture epistemic uncertainty, rendering them unable to express ignorance about unseen data distributions [61; 169]. To address this limitation, Bayesian Deep Learning (BDL) has emerged as a principled framework that places probability distributions over network parameters rather than learning fixed values. By inferring a posterior distribution over weights, BDL enables the model to express uncertainty regarding its own parameters, effectively capturing epistemic uncertainty through the variance of the predictive distribution [15; 4]. However, exact Bayesian inference is computationally intractable for high-dimensional deep networks, leading to the development of various approximation techniques. These include variational inference, which approximates the posterior with a simpler distribution, and the Laplace approximation, which utilizes the curvature of the loss function at the maximum a posteriori estimate [262; 77]. Despite their theoretical appeal, these methods often face challenges in scalability and calibration, with some studies indicating that standard approximations may underestimate uncertainty in data-scarce regions or fail to scale to modern architectures [88; 42].

Furthermore, recent investigations have revealed paradoxical behaviors in epistemic uncertainty estimation, such as the "epistemic uncertainty hole," where uncertainty metrics collapse in the presence of large models or specific training conditions, contradicting theoretical expectations that uncertainty should decrease with more data but remain sensitive to model complexity [96; 69]. These findings highlight that simply applying Bayesian machinery is insufficient; the quality of the posterior approximation and the choice of priors significantly impact the reliability of the resulting uncertainty estimates. Alternative approaches such as Deep Ensembles and Monte Carlo (MC) Dropout have gained popularity as scalable, albeit approximate, methods for uncertainty estimation. Deep Ensembles, which aggregate predictions from multiple independently trained networks, are often considered state-of-the-art for UQ, yet their theoretical connection to Bayesian inference has been a subject of debate until recent work demonstrated their equivalence to approximate Bayesian methods under specific assumptions [76; 239]. Similarly, MC Dropout interprets dropout layers during inference as a form of Bayesian approximation, though empirical evaluations suggest it may struggle to accurately reflect increased uncertainty in extrapolation regions compared to more rigorous Bayesian approaches [58; 15].

The distinction between aleatoric and epistemic uncertainty is not merely theoretical but has profound implications for practical applications ranging from air quality forecasting to medical imaging. In domains like air quality prediction, probabilistic models that account for both uncertainty types have demonstrated superior reliability compared to deterministic baselines, enabling more informed decision-making under volatile environmental conditions [268; 140]. In medical contexts, such as stroke analysis or wildfire danger forecasting, the ability to reject low-confidence predictions based on calibrated uncertainty thresholds can prevent erroneous diagnoses or inadequate emergency responses [202; 16]. However, challenges remain in ensuring that these uncertainty estimates are well-calibrated and robust to distribution shifts. Methods like Deep Evidential Regression and Credal Bayesian Deep Learning attempt to address these issues by leveraging evidence theory or imprecise probabilities to better disentangle uncertainty sources, though questions regarding their mathematical foundations and heuristic nature persist [284; 22; 253]. Ultimately, the

integration of robust UQ mechanisms is a prerequisite for the safe adoption of deep learning in mission-critical settings, requiring continued research into scalable, theoretically sound, and empirically reliable methods that can accurately quantify both what the model knows and what it does not [48; 32].

The Dual Nature of Predictive Uncertainty

The reliability of deep learning systems in high-stakes environments depends fundamentally on the rigorous distinction between two ontologically distinct sources of uncertainty: aleatoric and epistemic. Aleatoric uncertainty, often referred to as data or statistical uncertainty, arises from inherent noise within the data generation process, such as sensor errors, stochastic environmental factors, or irreducible randomness in the target variable [290; 55]. This form of uncertainty is theoretically irreducible; acquiring more data does not diminish the noise inherent to the phenomenon itself. Conversely, epistemic uncertainty, also known as model or systematic uncertainty, stems from a lack of knowledge, insufficient training data, or model misspecification [290; 57]. Unlike its aleatoric counterpart, epistemic uncertainty is reducible and should theoretically vanish as the model is exposed to more informative data or as the model architecture becomes sufficiently expressive to capture the underlying data distribution. While standard deterministic neural networks can be adapted to capture aleatoric uncertainty through heteroscedastic loss functions that predict variance alongside the mean, they inherently fail to account for epistemic uncertainty, often resulting in dangerously overconfident predictions when encountering out-of-distribution samples or data regimes far from the training distribution [284; 122]. Consequently, conflating these two distinct types can lead to misleading confidence intervals, particularly in dynamic systems where the distinction between irreducible noise and model ignorance is critical for safe decision-making and active learning strategies [169; 69].

Theoretical frameworks suggest that epistemic uncertainty can be quantified via the mutual information between model parameters and predictions, effectively measuring the reduction in entropy expected from acquiring new data [27; 86]. However, the practical realization of this dual nature is fraught with paradoxes that challenge standard assumptions. Recent empirical studies have identified an "epistemic uncertainty hole," a phenomenon where epistemic uncertainty estimates paradoxically collapse in the presence of large models or specific data regimes, directly contradicting the theoretical expectation that uncertainty should decrease with more data and increase with model complexity [96; 69]. This behavior undermines the applicative potential of BDL, particularly for out-of-distribution detection, where high epistemic uncertainty is the primary signal for model unreliability. Furthermore, common second-order uncertainty quantification methods often systematically blur bias-induced errors into aleatoric estimates, thereby underrepresenting the true extent of epistemic uncertainty and leading to a false sense of security regarding model knowledge [169]. This misattribution suggests that meaningful aleatoric estimates are feasible only if all relevant sources of epistemic uncertainty, including procedural variability and model bias, are properly represented and disentangled.

To address these limitations, researchers have proposed various architectural modifications and training strategies aimed at better disentangling these uncertainties. Methods such as cooperative Bayesian and variance networks aim to simultaneously learn aleatoric noise via mean variance estimation and epistemic uncertainty via Bayesian inference, thereby improving mean estimation while providing calibrated uncertainty bounds [79]. Additionally, innovative approaches like Hyper-Diffusion models and Epistemic Wrapping have been introduced to estimate both uncertainty types with single models or enhanced belief function posteriors, offering scalability without sacrificing accuracy [201; 131]. Despite these advancements, challenges remain in ensuring that these estimates are well-calibrated and robust to distributional shifts, as evidenced by studies showing that many methods fail to sufficiently increase uncertainty for samples far from the training distribution [20; 186]. The landscape is further complicated by the often-overlooked role of input uncertainty. Propagating uncertainty in the inputs through a neural network reveals that input noise can induce additional epistemic uncertainty at the output, necessitating unified formulations that account for input, data, and model uncertainties simultaneously [155; 45]. Research indicates that only specific approximate Bayesian methods, such as Ensembles and Flipout, are capable of explicitly modeling and propagating this input uncertainty, while others like MC-Dropout often fail to reflect increased input noise in their predictive uncertainty [122].

Moreover, the relationship between uncertainty measures and prediction accuracy is not linear or universal. The joint distribution of predictive and epistemic uncertainty reveals non-trivial patterns that depend heavily on the specific model architecture and dataset properties, meaning that marginalized uncertainty distributions alone cannot fully inform on prediction accuracy [27]. For instance, while epistemic uncertainty is theoretically expected to guide active learning by identifying regions where the model can learn most, empirical results show that this intuition does not always hold in high-dimensional computer vision domains, where epistemic uncertainty may not outperform predictive uncertainty for sample selection [27]. This complexity underscores the necessity of calibrating uncertainty measures for specific

architectures and data distributions rather than relying on generic heuristics. Ultimately, a nuanced understanding of the dual nature of predictive uncertainty, inclusive of input propagation, the potential for paradoxical behavior in large models, and the intricate dependence on architecture, is essential for developing trustworthy AI systems capable of recognizing their own limitations.

Limitations of Deterministic and Standard Approaches

Standard deterministic neural networks, despite achieving state-of-the-art performance across numerous domains, possess a fundamental architectural deficiency: they output point estimates that lack any inherent mechanism to quantify the confidence or reliability of their predictions. This absence of uncertainty awareness leads to a dangerous tendency toward overconfidence, particularly when models encounter out-of-distribution (OOD) data or inputs that deviate significantly from the training distribution [8]. In safety-critical applications such as autonomous driving and medical diagnosis, this overconfidence is not merely a theoretical shortcoming but a severe risk, as incorrect predictions made with high certainty can result in catastrophic outcomes [118]. The core of this issue lies in standard training procedures, which optimize exclusively for accuracy on the training set without imposing constraints to model uncertainty arising from data noise or model limitations [141]. Consequently, deterministic models frequently fail to distinguish between regions of high data density and extrapolation zones, assigning similarly high confidence levels to both known and unknown inputs [58].

To mitigate these deficiencies, researchers have explored probabilistic extensions, most notably heteroscedastic regression, where networks are trained to predict both the mean and variance of the target distribution. However, traditional implementations of this approach are fraught with significant pitfalls. A critical examination reveals that training neural networks to estimate the parameters of a heteroscedastic Gaussian distribution by maximizing the log-likelihood can lead to unstable and poor parameter estimates under specific conditions [114]. The reliance on gradient-based optimizers in conjunction with the log-likelihood loss function often causes the model to underestimate uncertainty or produce unreliable predictions, especially when the data contains outliers or when the model is misspecified [111]. In regression tasks, the common assumption of Gaussianity underlying mean-squared error loss restricts the hypothesis class; when this assumption is violated, the resulting aleatoric uncertainty estimates fail to capture the true noise structure of the data [111]. Furthermore, model misspecification does not only corrupt aleatoric uncertainty estimates; it also compromises epistemic uncertainty estimates in Bayesian treatments, leading to unreliable decision-making processes in downstream applications [111].

Even when moving beyond simple heteroscedastic models to more sophisticated approximate Bayesian methods, significant limitations persist. MC Dropout, widely adopted for its simplicity and computational efficiency, has been shown to struggle with accurately reflecting true uncertainty. Empirical investigations indicate that MC Dropout often fails to capture increased uncertainty in extrapolation and interpolation regions, behaving less reliably than traditional Bayesian approaches or Gaussian Processes in capturing both epistemic and aleatoric components [58]. While variations such as controlled dropout attempt to fix the number of dropout configurations to improve performance, the fundamental reliance on dropout as a proxy for Bayesian inference remains an approximation that can underestimate uncertainty, particularly under distribution shifts [281]. Similarly, deep ensembles, though often considered a strong baseline, rely on functional inconsistency among members which stems from unmanageable randomness in initialization and optimization; this inconsistency can collapse in specific cases, rendering the uncertainty estimates unreliable unless explicitly characterized and tuned against prior beliefs [239].

The challenge of disentangling and correctly estimating different types of uncertainty is further complicated by the behavior of modern overparameterized networks. In large models, epistemic uncertainty—the uncertainty reducible with more data—can paradoxically collapse even in the presence of limited training data, a phenomenon termed the "epistemic uncertainty hole" [96]. This collapse undermines the applicability of BDL for tasks like OOD detection, where high epistemic uncertainty is expected for unknown inputs. Additionally, current methods often fail to capture the full spectrum of epistemic uncertainty because they neglect model bias, systematically blurring bias-induced errors into aleatoric estimates and thereby underrepresenting the true model uncertainty [169]. The non-identifiability of parameters in overparameterized networks further complicates this, as substantial parameter uncertainty can persist even when the underlying function is identified, challenging the standard interpretation of epistemic uncertainty as purely reducible [287].

Furthermore, standard evaluation metrics themselves can be misleading when applied to models with significant model uncertainty. Metrics such as calibration error often conflate model uncertainty with aleatoric uncertainty, leading to incorrect conclusions about model reliability and potentially fostering trust in bad models while mistrusting good ones [36]. The inability of standard deterministic and even some probabilistic approaches to provide reliable, well-calibrated

confidence intervals without extensive modification or specific loss function engineering highlights a critical gap in current deep learning methodologies. This gap necessitates the development of more robust frameworks that can explicitly account for model misspecification, correctly disentangle uncertainty sources, and maintain reliability under distributional shifts, ensuring that uncertainty estimates are grounded in the empirical density of the input data rather than artifacts of the optimization process [177].

The Epistemological Necessity and Scope of UQ

The deployment of deep learning models in high-stakes environments necessitates a fundamental paradigm shift from prioritizing raw predictive accuracy to ensuring reliability through rigorous UQ. While standard deterministic neural networks frequently achieve state-of-the-art performance on benchmark datasets, they operate as opaque functions that often produce overconfident predictions when encountering adversarial examples, out-of-distribution samples, or subtle distributional shifts. This phenomenon poses severe risks in safety-critical domains where the cost of error is catastrophic, ranging from autonomous vehicle navigation to medical diagnosis and disaster response [8]. The epistemological argument for UQ rests on a crucial distinction: high accuracy on in-distribution data does not equate to robustness against the unknown. Consequently, UQ serves as the critical mechanism for identifying when a model should defer to human experts or reject a prediction entirely, thereby preventing fatal failures in applications where "I don't know" is a valid and necessary output [3] [258].

Theoretical frameworks categorize uncertainty into two primary sources, each demanding distinct mitigation strategies. Aleatoric uncertainty, or data uncertainty, arises from inherent noise in the observations, such as sensor noise, label ambiguity, or irreducible stochasticity in the physical process, and cannot be reduced by collecting more training data. In contrast, epistemic uncertainty, or model uncertainty, stems from a lack of knowledge regarding the optimal model parameters or architecture and is theoretically reducible with more representative data [8] [86]. Distinguishing between these sources is vital for practical decision-making; for instance, high aleatoric uncertainty in medical imaging might indicate ambiguous pathology requiring further diagnostic tests, whereas high epistemic uncertainty suggests the model is encountering a scenario outside its training distribution, necessitating model retraining or immediate human intervention [103]. However, recent analyses suggest that the relationship between these uncertainty types and actual prediction accuracy is non-trivial and highly dependent on model architecture and dataset properties, challenging the assumption that marginalized uncertainty measures alone are sufficient for reliability assessment [27].

The scope of this review extends across diverse domains where the consequences of uncertainty are profound and the failure of deterministic models is well-documented. In biomedical imaging, UQ is indispensable for tasks like MRI segmentation and disease detection, where models must quantify confidence to avoid hallucinating anatomical features or misdiagnosing conditions like COVID-19 [293] [199]. Research demonstrates that integrating segmentation uncertainty into downstream statistical analyses of imaging biomarkers significantly improves the fidelity of disease progression tracking, moving beyond point estimates to probabilistic assessments that reflect clinical reality [252]. Furthermore, the alignment of machine-generated uncertainty with human linguistic uncertainty in clinical reports remains a critical frontier, as current models often fail to capture the nuances of human expert hesitation, suggesting that true reliability requires bridging the gap between algorithmic confidence and clinical intuition [197]. Beyond medicine, UQ is critical in scientific simulations, such as inertial confinement fusion and proton dose calculation, where surrogate models must provide physically meaningful uncertainty bounds to ensure experimental validity and patient safety in radiotherapy [224] [260]. Similarly, in structural health monitoring and cosmological surveys, reliable uncertainty estimates enable the rejection of erroneous predictions and the robust handling of ambiguous data, such as overlapping galaxies or subtle structural cracks [Probabilistic segmentation of overlapping galaxies for large cosmological surveys] [167].

Despite these advances, significant gaps remain between theoretical potential and real-world deployment, forming a core component of this review's scope. A recurring challenge is the calibration of uncertainty estimates; many methods systematically underestimate predictive uncertainty, leading to overconfidence that can mislead decision-makers [103]. Moreover, the evaluation of UQ methods lacks standardization, with different metrics yielding conflicting assessments of method quality, highlighting the need for principled, frequentist coverage metrics to validate uncertainty claims [161] [171]. There is also a noted disconnect between machine-generated uncertainty and the variability inherent in human annotation, suggesting that current models do not fully capture the spectrum of plausible outcomes recognized by experts [Bayesian approaches for Quantifying Clinicians' Variability in Medical Image Quantification]. Addressing these discrepancies requires not only algorithmic improvements but also a deeper integration of domain-specific knowledge and rigorous benchmarking protocols to ensure that UQ mechanisms translate into trustworthy AI systems capable of operating safely in complex, dynamic environments [86]. The review further encompasses the evolution from computationally intractable exact Bayesian inference to scalable approximate techniques, such as efficient

posterior sampling for architectures like nnU-Net, which preserve performance while introducing robust uncertainty estimates [244]. By synthesizing these methodological advances with domain-specific challenges, this work aims to identify the critical barriers preventing the widespread adoption of uncertainty-aware deep learning in mission-critical applications.

Theoretical Foundations of Bayesian Deep Learning

The theoretical framework of BDL fundamentally reinterprets the architecture of neural networks by treating weights not as fixed deterministic values, but as random variables governed by probability distributions. This shift from a frequentist point estimate to a probabilistic formulation allows for the rigorous quantification of uncertainty, addressing critical limitations in standard deep learning such as overconfidence and poor calibration [61; 1]. In this paradigm, a Bayesian Neural Network (BNN) is defined by placing a prior distribution $p(\theta)$ over the network parameters θ , which represents beliefs about the model before observing data. Upon observing a dataset $\mathcal{D}$, Bayes' theorem is employed to compute the posterior distribution $p(\theta|\mathcal{D})$, combining the prior with the likelihood of the data [238; 13]. The predictive distribution for new inputs is obtained by marginalizing over this posterior, effectively averaging predictions across all plausible parameter configurations weighted by their posterior probability [1]. This marginalization process naturally decomposes predictive uncertainty into two distinct components: aleatoric uncertainty, which is inherent to the data generating process and irreducible, and epistemic uncertainty, which stems from model ignorance and is reducible with more data [22; 55].

A central theoretical challenge in BDL is the intractability of computing the exact posterior distribution due to the high dimensionality of the parameter space and the non-linear nature of deep neural networks [238; 61]. Consequently, the literature has developed various approximate inference techniques to navigate this complexity. Variational Inference (VI) approximates the true posterior with a tractable distribution by minimizing the Kullback-Leibler divergence, often utilizing the reparameterization trick to enable gradient-based optimization [13; 25]. Alternatively, sampling-based methods like Markov Chain Monte Carlo (MCMC) provide asymptotically exact samples but are often computationally prohibitive for large-scale models, prompting innovations such as Multilevel Monte Carlo methods applied to trace-class neural network priors to reduce computational complexity [Bayesian Deep Learning with Multilevel Trace-class Neural Networks; 238]. Another prominent approach involves the Laplace approximation, which approximates the posterior as a Gaussian centered at the maximum a posteriori (MAP) estimate, utilizing the curvature of the loss landscape [271; 233]. Recent developments have also explored subnetwork inference, demonstrating that performing Bayesian inference over a small subset of weights while keeping others as point estimates can preserve predictive uncertainty with significantly reduced computational cost [37].

Theoretical insights into the behavior of BNNs are significantly enhanced by analyzing the infinite-width limit of neural networks. In this regime, as the width of the hidden layers tends to infinity, the behavior of randomly initialized neural networks converges to that of Gaussian Processes (GPs) [23]. This duality provides a rigorous bridge between classical statistics and modern deep learning, allowing for exact closed-form solutions for predictive uncertainty in the infinite-width limit. Furthermore, the Neural Tangent Kernel (NTK) framework elucidates the training dynamics in this limit, revealing connections between gradient descent in wide networks and kernel regression [23; 172]. These theoretical results suggest that deep ensembles, often viewed as a non-Bayesian heuristic, can be interpreted as an approximate Bayesian method that mirrors the predictive variance of a GP in the infinite-width limit, thereby providing a unified perspective on uncertainty quantification [172; 76]. However, it is noted that finite ReLU BNNs can suffer from asymptotic overconfidence far from the training data, a pathology that can be corrected by extending finite networks with infinite features via the GP limit [278].

Despite these theoretical advances, significant debates persist regarding the practical efficacy and interpretation of uncertainty in BNNs. A notable phenomenon identified in recent literature is the "epistemic uncertainty hole," where epistemic uncertainty unexpectedly collapses in large models or with limited data, contradicting theoretical expectations that uncertainty should remain high in data-scarce regions [96]. This issue highlights the sensitivity of BNNs to the choice of priors and the approximations used during inference. The selection of priors is critical; while weight-space priors are common, they can be difficult to interpret and may lead to unintended inductive biases [134; 61]. To address prior misspecification and improve robustness, recent work has introduced Credal Bayesian Deep Learning, which utilizes sets of priors and likelihoods (credal sets) rather than single distributions, effectively training an infinite ensemble of BNNs to better distinguish between aleatoric and epistemic uncertainties [22]. Additionally, function-space variational inference has emerged as a powerful alternative to weight-space methods, allowing for the incorporation of meaningful prior information directly into the space of functions induced by the network, leading to more reliable uncertainty estimates [82; 134]. These developments collectively underscore the evolving theoretical landscape of

BDL, which seeks to balance computational tractability with the rigorous probabilistic foundations necessary for safe and reliable AI deployment.

Priors, Posteriors, and Inference Challenges

The formulation of BNNs is fundamentally constrained by the dual challenges of specifying appropriate prior distributions in high-dimensional spaces and computing the resulting posterior distributions amidst complex, non-convex likelihood landscapes. While the Bayesian paradigm offers a rigorous framework for UQ essential for large-scale AI, the practical implementation of this framework in deep learning reveals significant theoretical and computational fissures [42]. The choice of prior is particularly critical; standard practices often default to independent Gaussian priors over weights for computational convenience, yet these simplistic assumptions frequently fail to capture the structural properties of deep networks or induce necessary sparsity. Recent scholarship argues that more sophisticated priors are required to address overparameterization. For instance, the incorporation of spike-and-slab priors enables joint modeling of parameter and model uncertainty, effectively performing variable selection within the architecture [25]. Similarly, shrinkage priors such as the Horseshoe prior have been shown to provide stronger regularization and automatic variance specification compared to standard Gaussian mixtures, thereby mitigating the risks associated with ultra-high-dimensional weight spaces [25]. Beyond the weight space, there is a growing consensus that specifying priors directly in function space or predictive space offers greater interpretability and flexibility, allowing domain knowledge to be encoded more naturally than through opaque weight distributions [240; 134].

Despite the theoretical appeal of sophisticated priors, the primary bottleneck remains the intractability of computing the true posterior distribution. The likelihood landscapes of deep neural networks are characterized by high dimensionality, non-linearity, and multimodality, rendering traditional sampling-based methods like MCMC computationally prohibitive for large-scale models [14]. MCMC methods often struggle to explore the posterior surface efficiently within reasonable timeframes, failing to capture the full breadth of the distribution [238]. Consequently, the field has largely pivoted toward approximate inference techniques, with VI emerging as a dominant strategy due to its scalability via stochastic gradient descent. However, VI introduces its own biases; it often relies on mean-field assumptions that impose strong independence constraints across parameters, leading to posterior approximations that systematically underestimate uncertainty and fail to capture correlations between weights [178; 238]. Furthermore, the optimization landscape of the Evidence Lower Bound (ELBO) in VI can be difficult to navigate, sometimes resulting in solutions that concentrate around a single mode of the posterior, thereby missing the multimodal nature of the true distribution [14].

To mitigate the limitations of standard VI and the computational cost of MCMC, researchers have developed hybrid and alternative approximation strategies that seek to balance fidelity with feasibility. One prominent approach involves refining the Laplace approximation, which approximates the posterior as a Gaussian centered at the MAP estimate. While computationally efficient, the standard Laplace method can be overly confident; recent advancements address this by employing local linearization of activation functions or by constructing mixtures of Laplace approximations to better capture multimodality and improve uncertainty estimates in out-of-distribution regions [265; 90]. Another line of inquiry focuses on reducing the dimensionality of the inference problem. Methods such as subnetwork inference propose performing Bayesian inference only over a small, critical subset of weights while keeping the remainder as point estimates, thereby enabling the use of more expressive posterior approximations like full-covariance Gaussians on the selected subnetwork [37]. Similarly, active subspace methods identify low-dimensional directions in the parameter space that most significantly influence the output, allowing for scalable inference in these reduced dimensions [153].

The interplay between the choice of prior and the inference algorithm further complicates the landscape, occasionally revealing fundamental misspecifications in the BDL workflow. The "cold posterior" effect, where artificially sharpening the posterior yields better predictive performance, suggests a potential mismatch in either the prior, the likelihood, or the inference approximation itself [231]. This phenomenon highlights the delicate balance required in BNN design, where an inappropriate prior or a restrictive inference family can lead to miscalibrated uncertainty estimates. To robustify the inference process against such misspecification, recent work has explored the use of credal sets to represent sets of priors and likelihoods, aiming to better disentangle aleatoric and epistemic uncertainties [22]. Despite these advances, the tension between computational feasibility and the fidelity of the posterior approximation persists. While methods like deep ensembles have been shown to implicitly perform empirical Bayes with data-dependent priors, offering strong empirical performance, they lack the formal rigor of a fully specified Bayesian model with a fixed prior [182]. Ultimately, the challenge lies in developing inference techniques that can scale to

modern architectures while preserving the rich probabilistic structure necessary for reliable UQ, a goal that continues to drive innovation in approximate Bayesian computation [42; 61].

The Infinite Width Limit and Gaussian Processes

The theoretical analysis of neural networks in the limit of infinite width provides a rigorous mathematical foundation for understanding the behavior of deep learning models, particularly regarding UQ and generalization. As the width of a neural network tends to infinity, its behavior converges to that of a GP, a stochastic process where any finite collection of random variables has a multivariate normal distribution. This convergence allows for the derivation of exact analytical solutions for predictive uncertainty, bypassing the intractability often associated with finite-width BNNs. The connection between infinitely wide neural networks and GPs is formalized through the NTK, which describes the evolution of the network function during training under gradient descent. In this regime, the effect of random initialization and the training dynamics can be expressed with analytical tools, enabling researchers to obtain closed-form solutions for predictive uncertainty that are otherwise inaccessible in practical, finite-width settings [23].

The equivalence between wide neural networks and GPs offers a unique perspective on the nature of epistemic uncertainty. In the infinite-width limit, the posterior distribution over functions becomes a GP, allowing for precise characterization of uncertainty based on the distance from the training data. However, a critical distinction arises when considering specific activation functions, such as ReLU. While finite-width BNNs with ReLU activations can exhibit asymptotic overconfidence far from the training data—where the output variance grows only quadratically with distance—their infinite-width counterparts converge to a GP with a variance that grows cubically. This cubic growth ensures that the model remains maximally uncertain in regions far from the data, effectively fixing the issue of asymptotic overconfidence. Recent work has leveraged this insight to propose infinite-feature extensions for finite ReLU networks, combining the predictive power of standard BNNs near the data with the robust uncertainty properties of the infinite-width GP limit [278].

The NTK serves as the primary mechanism linking the training dynamics of wide neural networks to kernel methods. In the infinite-width limit, the NTK becomes constant during training, implying that the network behaves like a linear model in function space. This linearization facilitates the approximation of the posterior of a GP using the empirical NTK. Practical methods have been developed to exploit this connection, such as post-hoc sampling-based UQ techniques that construct efficient deep ensembles by employing gradient-descent sampling on linearized networks. These approaches demonstrate that approximating the GP posterior via the empirical NTK can yield state-of-the-art performance in uncertainty metrics while significantly reducing computational costs compared to traditional Bayesian methods [219]. Furthermore, the NTK framework has been instrumental in establishing theoretical equivalences between seemingly disparate methods. For instance, analysis within the NTK framework reveals that the uncertainty signal from Random Network Distillation (RND) is equivalent to the predictive variance of a deep ensemble in the infinite-width limit. By constructing specific target functions, RND can be shown to mirror the centered posterior predictive distribution of Bayesian inference, thereby unifying lightweight novelty detection methods with principled Bayesian frameworks [172].

Despite the theoretical elegance of the infinite-width limit, practical deep learning predominantly operates in the finite-width regime, where exact GP solutions do not directly apply. This discrepancy motivates the development of scalable approximations that capture the benefits of the infinite-width theory without incurring prohibitive computational costs. The transition from finite to infinite width also sheds light on the behavior of deep ensembles. Deep ensembles, often viewed as a frequentist approach, can be interpreted as an approximate Bayesian method that converges to a GP posterior as the number of ensemble members and network widths increase. Recent refinements propose explicitly characterizing the functional inconsistency among ensemble members via empirical covariance, effectively treating the ensemble as a GP to tune uncertainty with respect to training data and prior beliefs [239]. This perspective bridges the gap between the heuristic success of ensembles and the rigorous guarantees of Bayesian inference.

However, the application of infinite-width theories to real-world problems reveals certain limitations and nuances. While the infinite-width limit provides exact solutions, it assumes a specific scaling regime that may not reflect the behavior of modern over-parameterized models trained on finite datasets. For example, the contraction rates of epistemic uncertainty, which theoretically follow an $\mathcal{O}(1/N)$ rate in identifiable parametric models, may not hold in the over-parameterized regime, leading to unexplored behaviors where uncertainty does not vanish as expected even with large datasets [32]. Additionally, the assumption of linearity inherent in the NTK analysis may overlook the rich

feature learning dynamics present in finite-width networks, where the kernel itself evolves during training. Consequently, while the infinite-width limit and its connection to GPs provide invaluable theoretical insights into training dynamics and generalization, they serve primarily as a guiding principle for developing approximations rather than a direct substitute for finite-width modeling. The challenge remains to devise methods that retain the analytical tractability of the GP limit while accommodating the non-linear, feature-learning capabilities of finite-width deep networks [77].

Structural and Parameter Uncertainty

Within the theoretical foundations of BDL, a critical distinction exists between parameter uncertainty, which concerns the specific values of weights within a fixed architecture, and structural uncertainty, which involves ambiguity regarding the network architecture itself. While traditional BNNs place distributions over weights to capture epistemic uncertainty arising from limited data, they typically assume a fixed model structure. This assumption can lead to overconfident predictions if the chosen architecture is misspecified, as the model fails to account for the uncertainty inherent in the choice of depth, width, or connectivity patterns. Recent advancements emphasize that a comprehensive quantification of epistemic uncertainty requires inference in the joint space of model structures and parameters. [25] addresses this by proposing a scalable VI approach that incorporates model uncertainty through latent binary variables corresponding to the inclusion or exclusion of weights. By utilizing spike-and-slab priors, this method facilitates Bayesian Model Averaging (BMA), averaging predictions across multiple plausible structures rather than relying on a single "best" model, thereby providing a more rigorous probabilistic foundation than standard dropout techniques which primarily address parameter uncertainty.

The neglect of structural uncertainty contributes significantly to observed anomalies in UQ, such as the "epistemic uncertainty hole." [96] highlights a phenomenon where epistemic uncertainty metrics collapse in the presence of large models or limited training data, contradicting theoretical expectations. This collapse is partly attributed to the inability of standard approximate inference methods to adequately explore the posterior distribution over structures and parameters. Furthermore, in overparametrized networks, model parameters are often non-identifiable due to symmetries and redundant representations, meaning substantial parameter uncertainty can persist even when the underlying function is fully identified [287]. This complicates the disentanglement of uncertainty sources, as structural ambiguity can masquerade as parameter variance or vice versa. [169] argues that current methods often fail to capture critical components of epistemic uncertainty, particularly model bias, which is intrinsically linked to structural choices. When structural uncertainty is ignored, bias-induced errors are frequently blurred into aleatoric estimates, leading to misleadingly low epistemic uncertainty scores. [190] supports this by decomposing epistemic uncertainty into procedural variability and data variability, positing that structural uncertainty is a core component of procedural variability representing the uncertainty in the learning procedure and architectural path.

To mitigate these issues, various approximation strategies have been developed to handle the computational intractability of exact inference over structural spaces. [14] examines the performance of VI versus MCMC methods, finding that while MCMC offers asymptotic guarantees, it often struggles to explore the multimodal posteriors induced by structural variations in large networks. In contrast, VI can provide better UQ in practice, especially when combined with ensembling techniques. This aligns with the perspective that deep ensembles can be viewed as an approximate Bayesian method that effectively enlarges the epistemic part of the uncertainty, potentially capturing structural variations if the ensemble members differ in architecture [76]. However, explicit modeling of structural uncertainty remains superior for tasks requiring sparsity and interpretability, as demonstrated by methods that perform structural pruning through probabilistic reasoning rather than ad-hoc thresholding.

The impact of structural uncertainty extends to specific domains where the data structure itself is ambiguous. In graph neural networks, for instance, the graph topology introduces a layer of uncertainty distinct from node features. [144] proposes a unified approach to treat modeling errors and measurement uncertainty, acknowledging that ignoring the uncertainty in the relational structure can lead to severe miscalibration. Similarly, in scientific modeling, [229] demonstrates that prediction-oriented inference approaches provide better-calibrated uncertainties when model misspecification—often a result of incorrect structural assumptions—is present. These approaches adapt to the level of misspecification, offering a robust alternative to standard Bayesian inference which may underpredict uncertainty in the face of structural errors.

Ultimately, the calibration of epistemic uncertainty is deeply intertwined with the treatment of structural ambiguity. [69] identifies paradoxes where epistemic uncertainty measures violate fundamental principles, such as decreasing with more data, attributing these inconsistencies to poor posterior approximations that fail to capture structural fidelity. To address this, frameworks like Credal Bayesian Deep Learning utilize credal sets to train infinite ensembles of BNNs, offering robustness against prior and likelihood misspecification that often stems from rigid structural assumptions [22]. By acknowledging that the network architecture is not a fixed truth but a source of uncertainty itself, researchers can

develop models that are more robust and reliable. The choice of architecture and the method of inference are deeply intertwined, and ignoring structural uncertainty can lead to overconfident and potentially dangerous predictions in safety-critical applications [61]. Future research must continue to refine joint inference methods to balance computational feasibility with the rigorous quantification of all sources of epistemic uncertainty.

Approximate Inference Methodologies

The computational intractability of exact Bayesian inference in deep neural networks necessitates the adoption of approximate methodologies to estimate posterior distributions over weights and activations. As model complexity increases, the high dimensionality of the parameter space and the lack of identifiability render traditional MCMC methods prohibitively expensive and often unable to fully explore multimodal posteriors [14]. Consequently, the field has coalesced around three primary categories of approximation: VI, ensemble-based methods, and sampling-based approaches, each presenting distinct trade-offs between accuracy, computational cost, and scalability.

VI has emerged as a dominant strategy due to its superior computational complexity compared to sampling-based techniques. By framing inference as an optimization problem, VI approximates the intractable posterior with a simpler, tractable distribution, typically minimizing the Kullback-Leibler divergence or maximizing the ELBO [1]. However, standard mean-field VI often suffers from restrictive independence assumptions and a tendency to concentrate around a single mode, potentially leading to underestimation of uncertainty [178]. To address these limitations, recent innovations have introduced more expressive variational families. For instance, Boosted Bayesian Neural Networks utilize Boosting VI to iteratively construct a mixture of densities, thereby expanding the approximating family and improving generalization [21]. Similarly, Multiplicative Normalizing Flows have been employed to enhance the flexibility of the variational posterior, yielding predictive distributions that more closely mirror the true posterior and mitigating bias in parameter estimation [191]. Furthermore, function-space VI offers an alternative by inferring a posterior distribution directly over functions rather than parameters, allowing for the incorporation of meaningful prior information about the data-generating process [82]. Recent work also explores sparse subspace VI, which maintains a consistently highly sparse Bayesian model throughout training, significantly reducing FLOPs while preserving uncertainty metrics [203]. Additionally, large-scale score-based variational posterior inference has been proposed to avoid reparameterized sampling, enabling scalable inference on vision transformers and other large architectures [256].

Despite the efficiency of VI, ensemble-based methods, particularly Deep Ensembles, are frequently regarded as the state-of-the-art for UQ, often outperforming standard Bayesian approximations in calibration and robustness [76]. While originally proposed as a non-Bayesian technique, theoretical work has established that deep ensembles can be viewed as an approximate Bayesian method, effectively performing empirical Bayes with a data-dependent prior learned implicitly during training [182]. This perspective bridges the gap between the two paradigms, suggesting that the functional inconsistency among ensemble members serves as a proxy for epistemic uncertainty. Innovations in this domain include refining ensembles by explicitly characterizing functional inconsistency through empirical covariance to approximate a GP posterior, thereby enhancing reliability without significant computational overhead [239]. Additionally, hybrid approaches such as Mixtures of Laplace Approximations combine the benefits of ensembling and Bayesian approximation by predicting with a Gaussian mixture model composed of weighted Laplace approximations of independently trained networks, effectively mitigating overconfidence on outliers [90].

Sampling-based approaches, while theoretically grounded with asymptotic guarantees, face severe scalability challenges in deep learning contexts. Nevertheless, advancements aim to make these methods viable for large-scale architectures. Multilevel Monte Carlo (MLMC) methods, when combined with trace-class neural network priors, have been shown to reduce computational complexity to the canonical rate, making fully Bayesian inference feasible for specific model classes [Bayesian Deep Learning with Multilevel Trace-class Neural Networks]. Similarly, affine invariant ensemble transform methods, derived from the ensemble Kalman filter, provide efficient interacting particle systems to sample from approximate posteriors in logistic regression and neural network settings [250]. In contrast to full network sampling, subnetwork inference proposes performing Bayesian inference only over a small subset of model weights while keeping others as point estimates. This approach leverages the overparameterization of neural networks to maintain accurate predictive posteriors with significantly reduced computational burden, often comparing favorably to full-network approximations [37].

The interplay between optimization and inference further complicates the landscape. The generalized Gauss-Newton method, often used in conjunction with Laplace and Gaussian variational approximations, simplifies the underlying probabilistic model locally into a linear or GP model, clarifying the impact of optimization choices on posterior approximation [271]. Moreover, recent critiques highlight that common approximations like MC Dropout may fail to reliably capture increased uncertainty in extrapolation regions compared to more rigorous Bayesian approaches,

underscoring the need for careful method selection based on the specific requirements of the application [58]. Hybrid strategies continue to evolve, such as combining VI with stochastic depth or utilizing score-based objectives to avoid reparameterized sampling, thereby enabling scalable inference on vision transformers and other large architectures [280; 256]. Ultimately, the choice of inference methodology involves a nuanced balance where no single approach universally dominates; rather, the optimal strategy depends on the specific architectural constraints, the nature of the uncertainty required (aleatoric vs. epistemic), and the available computational resources [48].

Variational Inference and Mean-Field Approximations

VI has become the dominant paradigm for scalable BDL, primarily because it transforms the intractable problem of posterior integration into a tractable optimization problem by minimizing the KL divergence between a variational distribution and the true posterior [13]. While standard implementations rely on mean-field approximations that assume independence among weight parameters to facilitate efficient gradient-based updates, this restrictive factorization often leads to a systematic underestimation of uncertainty. Specifically, mean-field VI tends to concentrate around a single mode of the posterior, failing to capture the complex, multimodal structures inherent in deep neural network loss landscapes [14]. To overcome these expressiveness limitations without sacrificing computational feasibility, recent research has diverged into three primary streams: constructing richer variational families, enforcing sparsity to reduce dimensionality, and shifting the locus of inference to lower-dimensional spaces.

To address the inability of single-density approximations to capture multimodality, researchers have developed Boosted VI (BVI), which iteratively constructs a mixture of densities to approximate the posterior. This approach, exemplified by Boosted Bayesian Neural Networks (BBNN), significantly enhances the expressiveness of the variational family, leading to improved generalization and more accurate uncertainty estimates compared to standard VI, albeit with increased computational complexity [21]. Parallel to mixing densities, alternative divergence measures have been proposed to replace the standard KL term in the ELBO. By substituting the KL divergence with a Maximum Mean Discrepancy (MMD) estimator, models become more robust to prior misspecification and exhibit superior calibration, while also demonstrating enhanced resistance to adversarial attacks [142]. Furthermore, score-based VI methods have emerged to bypass the need for reparameterized sampling. These methods combine score matching losses with proximal penalties, enabling scalable training on large-scale architectures like Vision Transformers and supporting richer variational density families that avoid the constraints of standard reparameterization tricks [256].

A second major research thrust focuses on mitigating the computational burden of VI through sparsity and subspace learning, addressing the prohibitive cost of maintaining distributions over millions of parameters. Sparse Subspace VI (SSVI) introduces a framework that learns within a consistently low-dimensional sparse subspace throughout both training and inference. By alternating between basis selection and parameter optimization, SSVI achieves up to a 20-fold reduction in FLOPs during training and significant model compression, often surpassing dense VI models in both accuracy and uncertainty metrics [203]. Complementary to this, other approaches propose learning active subspaces by identifying the specific parameter directions that most significantly influence the network output, thereby enabling effective Bayesian inference in a reduced dimensionality setting [153]. Structural uncertainty has also been explicitly modeled using spike-and-slab priors within a VI framework, allowing for joint inference over model structures and parameters. This results in highly sparse networks that maintain predictive accuracy while providing a principled mechanism for model selection [25]. Additionally, sparsity has been induced through source-specific uncertainty frameworks and shrinkage priors, offering new mechanisms for regularization and efficient model design [236].

Recognizing that the high dimensionality of the weight space is a primary bottleneck, recent advancements have shifted the locus of inference to more manageable function or output spaces. VI in the Final layer Output space (VIFO) leverages the significantly smaller dimensionality of the output space compared to the weight space, offering a favorable trade-off between runtime and UQ, particularly for out-of-distribution data [220]. Function-space VI methods further decouple inference from specific architectural choices by placing priors directly on the functions induced by the network. This approach facilitates the incorporation of meaningful prior information about the data-generating process and has demonstrated state-of-the-art performance in safety-critical tasks where reliable uncertainty estimation is paramount [82]. This perspective is reinforced by findings that deep ensembles can be viewed as approximating a GP posterior in function space, effectively bridging the gap between ensemble methods and rigorous Bayesian inference [239].

Despite these methodological advancements, challenges regarding calibration and the "cold posterior" effect persist, indicating that optimization dynamics and likelihood misspecification remain critical factors. Studies suggest that standard VI often requires tempering the posterior to achieve optimal predictive performance, pointing to potential issues with overly strong priors or likelihood assumptions [231]. To address calibration directly, frameworks such as Calibration-Aware Bayesian Neural Networks (CA-BNNs) integrate data-dependent regularizers with variational

objectives to improve reliability [138]. Moreover, sampling-free approaches utilizing multiplicative activation noise or local linearization have been developed to provide efficient uncertainty estimates without the stochasticity inherent in traditional sampling-based VI, proving effective in safety-critical domains [126] [265]. The interplay between optimization algorithms and inference is also critical; for instance, Variational Laplace exploits local curvature information to estimate the ELBO without stochastic sampling, often outperforming standard VI in calibration metrics while avoiding the convergence issues associated with variance parameters in standard implementations [233]. Collectively, these diverse methodologies illustrate a field evolving from simple mean-field approximations toward sophisticated, scalable, and expressive frameworks capable of handling the complexities of modern deep learning applications.

Monte Carlo Sampling and Dropout Approximations

MCMC methods represent the theoretical gold standard for approximate inference in BDL, offering asymptotically exact samples from the posterior distribution. However, the computational burden of exploring high-dimensional, multimodal posteriors in modern deep architectures often renders MCMC prohibitive for practical applications [14; 238]. To address this scalability bottleneck, MC Dropout was introduced as a pragmatic variational approximation that interprets the activation of dropout layers at test time as a form of Bayesian model averaging. This approach enables the estimation of epistemic uncertainty without requiring architectural changes or the extensive retraining associated with ensemble methods, securing its status as a widely adopted heuristic for UQ [152; 43]. Despite its popularity, the theoretical grounding of MC Dropout remains contentious, with growing evidence suggesting that its underlying mean-field approximation is often too restrictive to capture the full complexity of the true posterior, leading to significant discrepancies between estimated and actual uncertainty.

A primary limitation of standard MC Dropout is its failure to adhere to fundamental Bayesian principles regarding the reduction of epistemic uncertainty as training data increases. Empirical investigations have demonstrated that MC Dropout often struggles to accurately reflect true uncertainty, particularly failing to exhibit increased uncertainty in extrapolation and interpolation regions where rigorous Bayesian models typically excel [58]. This phenomenon, described as the "epistemic uncertainty hole," indicates that the variational family implicit in dropout may collapse in the presence of large models or specific data regimes, resulting in overconfident predictions even in data-scarce regions or when facing out-of-distribution inputs [96; 123]. Furthermore, comparative studies reveal that while MC Dropout is computationally efficient, it frequently yields uncertainty estimates of inferior quality compared to Deep Ensembles or bootstrapped neural networks, which better approximate the gold standard provided by MCMC in both classification and regression tasks [161; 48]. The method also exhibits architectural sensitivity, where the quality of uncertainty estimates varies significantly based on design choices, necessitating careful tuning to avoid unreliable results in safety-critical domains like medical imaging or autonomous driving [152; 199]. Additionally, recent findings suggest that MC Dropout may fail to properly model input uncertainty, whereas methods like Ensembles and Flipout demonstrate a superior capacity to propagate input noise into predictive uncertainty [122].

To mitigate the computational inefficiencies and variance issues inherent in standard Monte Carlo sampling via dropout, researchers have proposed advanced sampling frameworks such as MLMC. By treating dropout masks as a source of epistemic randomness and constructing coupled coarse-fine estimators, MLMC frameworks can significantly reduce sampling variance at a fixed evaluation budget, offering a more efficient route to UQ than single-level MC Dropout [274]. This approach leverages a hierarchy of fidelities defined by the number of stochastic forward passes, utilizing telescoping estimators to maintain unbiasedness while optimizing the cost-accuracy trade-off. Similarly, innovations like Controlled Dropout aim to fix the number of dropout configurations to stabilize performance metrics, addressing the stochastic volatility that can plague standard implementations [281]. Other works explore modifying the objective function itself, such as the introduction of the Second-Moment Loss, which explicitly optimizes model variance alongside the mean to counteract the tendency of standard MC Dropout to underestimate uncertainty [222]. In the context of modern residual architectures, Monte Carlo Stochastic Depth has emerged as a robust alternative, providing a theoretically grounded connection to VI while achieving competitive calibration and uncertainty ranking in object detection tasks [280].

Despite these methodological advancements, the consensus in the literature indicates that while MC Dropout serves as a valuable heuristic for scalable UQ, it often lacks the theoretical robustness of full Bayesian inference or the empirical reliability of deep ensembles, particularly in scenarios requiring precise calibration or the disentanglement of uncertainty types [32; 42]. Applications in fields ranging from rainfall-runoff modeling to customer lifetime value prediction have shown that while MC Dropout can provide reasonable estimates, it often requires specific architectural adaptations or hybridization with ensemble techniques to achieve well-calibrated results [272; 53]. Consequently, the choice of inference methodology involves a nuanced trade-off between the computational tractability of dropout-based approximations and the statistical fidelity of more expensive sampling or ensemble strategies, with the latter often preferred when the cost of uncertainty miscalibration is high [185; 182].

Deep Ensembles and Non-Bayesian Approaches

Within the spectrum of approximate inference methodologies, Deep Ensembles (DE) have emerged as a dominant non-Bayesian baseline, frequently outperforming single BNNs in both predictive accuracy and calibration. Originally conceptualized as a frequentist technique involving the training of multiple independent networks with distinct random initializations, DEs effectively capture multimodality in the posterior distribution, providing robust estimates of epistemic uncertainty through the disagreement among ensemble members [76]. While initially viewed as an ad-hoc procedure distinct from principled Bayesian inference, recent theoretical advancements have recontextualized deep ensembles as methods that implicitly perform empirical Bayes procedures. Specifically, it has been demonstrated that deep ensembles execute exact Bayesian averaging using a posterior derived from an implicitly learned, data-dependent prior, which often takes the form of a mixture of point masses [182]. This perspective resolves the apparent contradiction between the "principled" nature of BNNs and the superior empirical performance of ensembles, suggesting that the implicit prior learned by ensembles is often better suited to the data than the fixed priors assumed in standard BNN formulations.

The relationship between deep ensembles and other probabilistic frameworks has been further elucidated through connections to GPs and the NTK. In the limit of infinite network width, the uncertainty signal generated by deep ensembles—specifically the predictive variance—has been shown to be equivalent to that of certain Bayesian inference methods and even RND techniques [172]. However, standard ensembles rely on unmanageable randomness from initialization and stochastic gradient descent, which can lead to a collapse of functional inconsistency in specific cases. To address this, researchers have proposed refining standard ensembles by explicitly characterizing the functional inconsistency among members via empirical covariance, thereby defining a GP that approximates the Bayesian posterior more reliably than unmanaged randomness [239]. This approach, often referred to as DE-GP, allows for the tuning of both mean and covariance functions with respect to training data under functional VI paradigms, handling classification tasks directly without the need for regression proxies. Such developments highlight that while standard ensembles are effective, structured variations can enforce diversity in function space to prevent the collapse of uncertainty estimates, particularly in physics-informed contexts where repulsive ensembles are used to maintain diversity [286].

Despite their strengths, deep ensembles are not without limitations, particularly regarding the calibration of epistemic uncertainty and the disentanglement of uncertainty sources. A critical issue identified in the literature is the "epistemic uncertainty hole," where epistemic uncertainty metrics paradoxically collapse in the presence of large models or small datasets, contradicting theoretical expectations that uncertainty should decrease with more data and increase with model complexity [96]. Experimental studies have revealed that common measures of epistemic uncertainty in deep ensembles often violate these fundamental principles, necessitating regularization strategies such as "conflictual loss" to restore the expected behavior of uncertainty metrics [69]. Additionally, while ensembles are effective at capturing total predictive uncertainty, they do not inherently disentangle aleatoric (data) and epistemic (model) uncertainties as cleanly as some dedicated Bayesian approaches. Comparative studies in domains ranging from interatomic potentials to weather forecasting indicate that while ensembles generally provide conservative uncertainty estimates, methods like VI or specific hybrid frameworks may offer superior disentanglement capabilities required for active learning or safety-critical decision-making [BAYESIAN NEURAL NETWORKS VERSUS DEEP ENSEMBLES FOR UNCERTAINTY QUANTIFICATION IN MACHINE LEARNING INTERATOMIC POTENTIALS; BRIDGING THE GAP BETWEEN BAYESIAN DEEP LEARNING AND ENSEMBLE WEATHER FORECASTS].

To address these calibration and disentanglement challenges, several hybrid and post-hoc methods have been developed that leverage the efficiency of ensembles while incorporating Bayesian rigor. For instance, combining deep ensembles with Laplace approximations allows for the construction of Gaussian mixture model posteriors that mitigate overconfidence on outliers with minimal computational overhead [90]. Similarly, the integration of Bayesian Optimization with deep ensembles (BODE) has been shown to significantly reduce both aleatoric and epistemic uncertainty components in system safety applications, outperforming manually tuned baselines [94]. However, practitioners must remain cautious, as the number of networks in an ensemble can influence uncertainty estimation trends; increasing the ensemble size without post-hoc calibration can lead to an underestimation of uncertainty, potentially causing unintended exploratory behavior in downstream tasks like Bayesian optimization [159].

Furthermore, while ensembles are often viewed as the gold standard, recent work suggests that single-model approaches, such as Hyper-Diffusion models or Epistemic Neural Networks, can achieve comparable or superior uncertainty estimation with orders of magnitude less computation, challenging the necessity of large ensembles for all applications [Estimating Epistemic and Aleatoric Uncertainty with a Single Model; Epistemic Neural Networks]. Ultimately, the choice between deep ensembles and other approximate inference methods depends on the specific requirements for computational efficiency, the need for strict uncertainty disentanglement, and the criticality of calibration in the target domain.

Efficient and Hybrid Inference Strategies

The computational intractability of performing full Bayesian inference over the vast parameter spaces of modern deep neural networks has necessitated the development of efficient and hybrid approximation strategies. These approaches seek to retain the robust UQ capabilities of BNNs while mitigating the prohibitive costs associated with sampling or optimizing distributions over every weight in the network. A primary strategy involves restricting Bayesian inference to specific subsets of the network architecture, most notably the final layer, while keeping the preceding feature extraction layers deterministic. This hybrid philosophy leverages the empirical observation that pre-trained deep networks often learn robust feature representations that do not require probabilistic modeling to be effective, whereas uncertainty is most critical in the final decision-making stage. Research indicates that linearizing a deep network around the parameters of the final connected layer to form a Bayesian Generalized Linear Model yields UQ performance comparable to full-network linearization, but with substantially improved computational efficiency [116]. Building on this, the Bayesian Last Layer (BLL) method assumes distributed weights only in the linear output layer and allows for efficient training via backpropagation by reformulating the log-marginal likelihood, thereby achieving high log-predictive density on test data compared to full VI methods [235]. Extending this concept further, VIFO performs VI directly in the output space rather than the parameter space; since the output space is significantly smaller, VIFO offers a favorable trade-off between runtime and uncertainty quality, particularly for out-of-distribution data [220].

Beyond layer-specific restrictions, recent works have introduced adaptable frameworks that transform standard pre-trained models into BNNs with minimal overhead, addressing the instability often associated with training large BNNs from scratch. The Adaptable Bayesian Neural Network (ABNN) strategy attaches simple BNN adaptation layers to normalization layers of pre-trained models, requiring only a few fine-tuning steps to enhance UQ without the computational budget typically associated with ensemble methods [102]. This post-hoc philosophy is further exemplified in medical imaging, where efficient schemes for posterior sampling of weight space have been developed to estimate uncertainty for nnU-Net architectures without modifying their original, highly optimized structure [244]. These methods align with the broader philosophy of hybrid models, where the bulk of the network remains deterministic to preserve predictive accuracy, while stochasticity is introduced judiciously to capture epistemic uncertainty. Furthermore, subnetwork inference frameworks demonstrate that performing inference over a small, carefully selected subset of model weights-while keeping the rest as point estimates-can accurately represent the posterior predictive distribution of the full network. This enables the use of expressive posterior approximations like linearized Laplace on manageable scales, comparing favorably to ensembles and less expressive posterior approximations over full networks [37]. Complementary to this, SSVI introduces a fully sparse BNN framework that maintains a consistently highly sparse Bayesian model throughout training and inference, achieving significant compression and FLOPs reduction while maintaining or improving uncertainty metrics [203].

Other efficient strategies exploit the structural properties of neural networks or introduce novel regularization mechanisms to estimate uncertainty without explicit sampling over weights. One such approach involves probabilistic reasoning over the depth of the network. By exploiting the sequential structure of feed-forward networks, methods can evaluate training objectives and make predictions with a single forward pass, marginalizing over different depths to yield model uncertainty without the need for multiple stochastic forward passes typical of MC Dropout [7]. This concept is extended to residual networks, where training using stochastic depth can be interpreted as a VI approximation to the intractable posterior over weights, producing well-calibrated probabilities with only minor structural changes [137]. To address limitations in standard dropout and structural heuristics, VI approaches adapted for model uncertainty allow for inference in the joint space of structures and parameters. By utilizing spike-and-slab priors to induce sparsity, these methods perform rigorous Bayesian model selection or averaging, providing a more probabilistic foundation for pruning compared to heuristic dropout methods [25]. Additionally, variation due to regularization (RegVar) has emerged as a tractable method for large networks; by deterministically tweaking the training loss during fine-tuning, this approach recovers the Laplace approximation variance without explicitly computing the Hessian, thus scaling effectively with network size while providing rigorous uncertainty estimates [77].

The landscape of efficient inference also includes methods that integrate implicit uncertainty through input augmentation or cooperative architectures to avoid the impractical inference over millions of parameters. Implicit Bayesian neural networks propose augmenting each layer's inputs with latent variables to induce a distribution

capturing uncertainty, maintaining state-of-the-art calibration on large-scale tasks without the cost of weight-space inference [174]. In scenarios where disentangling uncertainty types is critical, cooperative training of a variance network with a BNN has been shown to effectively separate aleatoric and epistemic uncertainties while improving mean estimation, offering a robust and scalable alternative to difficult-to-train standalone BNNs [79]. Despite the promise of these hybrid and efficient methods, challenges remain regarding the mathematical grounding of some single-model approaches. While methods like Deep Evidential Regression claim to disentangle uncertainties efficiently, theoretical analyses suggest they may be heuristic rather than exact, prompting continued refinement of how these efficient methods mathematically ground their uncertainty estimates [284]. Nevertheless, the consensus across recent literature suggests that hybrid strategies, whether through last-layer inference, subnetwork selection, structural exploitation, or regularization-based approximations, offer a pragmatic path toward deploying uncertainty-aware deep learning in resource-constrained and safety-critical applications.

Disentangling and Quantifying Uncertainty Types

The rigorous separation of aleatoric and epistemic uncertainty is a foundational requirement for deploying machine learning models in safety-critical domains, yet achieving this disentanglement remains a complex theoretical and practical challenge. Aleatoric uncertainty, representing irreducible noise inherent in the data generation process, is typically modeled by having the network output parameters of a likelihood distribution, such as the mean and variance of a heteroscedastic Gaussian [55]. Conversely, epistemic uncertainty, which stems from a lack of knowledge about the model parameters and is reducible with more data, typically requires Bayesian inference or ensemble methods to quantify the spread of the posterior distribution over weights [27]. While Bayesian frameworks theoretically provide a clear distinction via the posterior parameter distribution and the likelihood function, practical implementations often struggle to maintain this separation, leading to metrics that conflate the two sources or fail to capture the full spectrum of model ignorance [169].

A primary strategy for isolating epistemic uncertainty involves the use of model ensembles or approximate Bayesian posteriors. Deep ensembles, consisting of independently trained networks, have emerged as a state-of-the-art non-Bayesian technique that effectively captures model disagreement as a proxy for epistemic uncertainty [76]. Recent work has further refined this approach by viewing deep ensembles through a Gaussian Process lens, explicitly characterizing functional inconsistency to prevent the collapse of uncertainty estimates in specific regimes [239]. Similarly, Monte Carlo dropout offers a computationally efficient approximation to Bayesian inference by sampling sub-models at inference time; however, empirical studies suggest that this method can be unreliable, particularly failing to reflect increased uncertainty in extrapolation regions compared to more rigorous Bayesian approaches or Gaussian Processes [58]. To address the computational intractability of large ensembles, novel architectures like the Epistemic Neural Network (ENN) and Hyper-diffusion models have been introduced, enabling the estimation of joint predictions and both uncertainty types with a single model or minimal incremental computation [127; 201].

Despite these methodological advances, significant contradictions exist regarding the behavior and calibration of these uncertainty estimates. Theoretical expectations dictate that epistemic uncertainty should decrease as the training dataset grows and increase with model complexity; however, experimental evidence reveals an "epistemic uncertainty hole" where uncertainty collapses in the presence of large models or even small datasets, contradicting fundamental Bayesian principles [96]. This phenomenon is further exacerbated by poor posterior approximations, leading to scenarios where common calibration techniques fail to satisfy basic data-related and model-related principles of epistemic uncertainty [69]. Furthermore, the interaction between uncertainty types is intricate; prediction accuracy depends on the joint distribution of aleatoric and epistemic uncertainties in ways that marginalized measures cannot capture, suggesting that simple decomposition may not always improve downstream tasks like active learning or medical diagnosis without careful calibration to the specific architecture and dataset [27].

Challenges also persist in the accurate modeling of aleatoric uncertainty, particularly when dealing with heteroscedastic noise or input uncertainty. Standard approaches that maximize log-likelihood to learn heteroscedastic variance can lead to poor parameter estimates due to optimization hazards, necessitating alternative loss formulations like β -NLL to ensure robustness [114]. Moreover, most methods implicitly assume perfect inputs, ignoring the propagation of input noise into predictive uncertainty. Research indicates that only specific methods, such as ensembles and certain Bayesian approximations like Flipout, can effectively model and propagate explicit input uncertainty, whereas others remain overconfident despite noisy inputs [122; 155]. The failure to account for model bias further complicates this landscape, as high bias can lead to misleadingly low epistemic estimates while blurring bias-induced errors into aleatoric components, thereby undermining the validity of the disentanglement [169].

To rigorously assess the quality of these disentangled estimates, new metrics and frameworks are being developed. The concept of "Disentanglement Error" has been proposed to benchmark how well different methods isolate uncertainty types, with findings indicating that while deep ensembles with information-theoretic measures perform well, there is still considerable room for improvement [241]. Additionally, axiomatic assessments and proper scoring rules are being utilized to provide a principled foundation for evaluating uncertainty in regression settings, moving beyond heuristic validations to ensure that uncertainty measures align with desired properties such as tail sensitivity and out-of-distribution responsiveness [207; 279]. These developments highlight that while the distinction between aleatoric and epistemic uncertainty is conceptually clear, realizing this separation in practice requires addressing issues of

calibration, model bias, input noise propagation, and the development of more robust evaluation metrics that go beyond simple entropy or variance calculations.

Modeling Aleatoric Uncertainty

Aleatoric uncertainty, intrinsic to the data generation process, represents the irreducible noise that persists regardless of the volume of training data available. In deep learning, this form of uncertainty is frequently characterized by heteroscedasticity, where the noise level is not constant but varies as a function of the input. To capture this input-dependent variability, contemporary approaches typically train neural networks to predict the parameters of an output distribution, such as the mean and variance of a Gaussian, rather than producing single point estimates [8]. This paradigm allows models to adapt their confidence based on the ambiguity of specific inputs, a capability crucial for tasks like regression and image segmentation where measurement errors or label noise fluctuate across the domain [Modeling Aleatoric Uncertainty]. However, the theoretical clarity of this approach often obscures significant practical challenges in implementation and interpretation, particularly regarding optimization stability and the fidelity of distributional assumptions.

A primary difficulty lies in the optimization landscape of standard probabilistic formulations. The common practice of maximizing the log-likelihood of observed data under a predicted heteroscedastic Gaussian distribution can lead to unstable training dynamics and poor parameter estimates. Research indicates that gradient-based optimizers may converge to solutions where the predicted variance collapses or becomes unreasonably large for outliers, failing to reflect true data noise [114]. To mitigate these hazards, alternative loss formulations such as β -NLL have been proposed, which weight the contribution of each data point by its estimated variance, thereby improving robustness and predictive performance across diverse domains [114]. Furthermore, the choice of distributional assumption is critical; while Gaussian assumptions are dominant, they may be ill-suited for data with heavy-tailed noise. Approaches utilizing Auxiliary Uncertainty Estimators (AuxUE) have demonstrated that approximating prediction errors with a Laplace distribution can yield more robust aleatoric estimates in vision regression tasks compared to standard Gaussian models [186].

The accuracy of aleatoric modeling is deeply intertwined with the correctness of the underlying model specification. When the assumed likelihood function does not match the true data-generating process, the resulting uncertainty estimates become unreliable. Specifically, under model misspecification, the model may incorrectly absorb structural bias into the aleatoric component, leading to an overestimation of data noise and a concurrent underestimation of epistemic uncertainty [111]. This misattribution is exacerbated by the fact that high model bias can systematically blur the distinction between reducible and irreducible uncertainty, causing second-order methods to report misleadingly low epistemic uncertainty while inflating aleatoric estimates [169]. Consequently, meaningful disentanglement of uncertainty types is contingent upon the model's ability to accurately represent the true conditional distribution of the targets.

Complicating this landscape further is the often-overlooked role of input uncertainty. Traditional frameworks typically assume inputs are noise-free, yet in real-world scenarios, sensor noise and measurement errors introduce uncertainty at the source. Propagating this input uncertainty through the network is essential for a complete decomposition of predictive uncertainty. Methods that explicitly incorporate input noise, such as Errors-in-Variables models in Bayesian deep regression, demonstrate that accounting for noisy inputs leads to a more statistically consistent separation of aleatoric and epistemic components [45]. Moreover, unified formulations suggest that propagating input uncertainty can induce epistemic-like behavior at the output, blurring the boundaries between uncertainty sources and necessitating architectural designs that simultaneously estimate input, data, and model uncertainty [155].

Empirical evaluations reveal that even when aleatoric uncertainty is explicitly modeled, the resulting estimates often suffer from miscalibration. Comparative studies of methods like Deep Ensembles and Deep Evidential Regression show that while predicted aleatoric uncertainty generally scales with injected noise levels, it frequently fails to align quantitatively with the true standard deviation of the noise, particularly in high-dimensional settings or under high-noise conditions [120]. In safety-critical domains like medical imaging, predictive uncertainty is systematically underestimated, requiring post-hoc calibration techniques such as σ -scaling to ensure that uncertainty intervals reliably cover ground truth values [103]. Additionally, standard metrics may fail to detect mismatches in error distributions; statistical tests based on Mahalanobis distance have shown that many models fail to capture heavy-tailed empirical errors, highlighting the need for uncertainty measures that appropriately upper-bound realistic error distributions [269]. Ultimately, accurate modeling of aleatoric uncertainty requires not only flexible distributional

assumptions and robust loss functions but also rigorous calibration and a nuanced understanding of how input noise and model bias interact to shape the predictive landscape.

Capturing Epistemic Uncertainty

Epistemic uncertainty, fundamentally defined as the reducible uncertainty arising from a model's lack of knowledge or insufficient training data, is distinct from aleatoric uncertainty, which represents inherent noise in the data generation process. While this distinction is widely accepted, capturing epistemic uncertainty robustly remains a significant challenge due to the complex interplay between model bias, hypothesis space selection, and estimator quality. [290] provides a rigorous taxonomy, distinguishing between model uncertainty stemming from the choice of hypothesis space and approximation uncertainty reflecting the quality of the estimator relative to the best hypothesis. This granularity is critical because, as argued in [169], many current methods based on second-order distributions systematically underestimate epistemic uncertainty by failing to account for model bias. Consequently, bias-induced errors are often erroneously attributed to aleatoric noise, leading to misleadingly low estimates of reducible uncertainty and compromising the model's ability to identify out-of-distribution samples.

Theoretical expectations dictate that epistemic uncertainty should decrease as the training dataset grows and increase with model complexity due to the curse of dimensionality. However, empirical evidence suggests that standard approaches frequently violate these principles. [69] highlights a paradox where measures of epistemic uncertainty in Deep Ensembles and Bayesian Neural Networks often fail to increase with model expressiveness and do not consistently decrease with larger datasets. This phenomenon, termed the "epistemic uncertainty hole" in [96], describes a collapse of uncertainty metrics in large models, effectively undermining their utility for safety-critical applications like out-of-distribution detection. To address these calibration failures, recent work proposes novel regularization techniques, such as conflictual loss, designed to enforce the expected behavioral properties of epistemic uncertainty without sacrificing predictive performance.

In response to the computational intractability of exact Bayesian inference and the limitations of standard approximations like Monte Carlo Dropout, researchers are developing more efficient and theoretically grounded strategies. [213] introduces a frequentist approach where the model is conditioned on its own previous outputs; the covariance between sequential predictions serves as a rigorous metric for epistemic uncertainty, bypassing the need for weight distributions. Similarly, [127] proposes the "epinet," an architecture that supplements conventional neural networks to estimate uncertainty with modest incremental computation, claiming to outperform large ensembles while maintaining efficiency. In the realm of generative modeling, [201] presents Hyper-Diffusion models, which leverage conditional diffusion to estimate both uncertainty types using a single model, resolving the scalability issues inherent in traditional ensemble methods. These single-model approaches aim to preserve the ability to distinguish between reducible and irreducible uncertainty while avoiding the heavy computational burden of full Bayesian inference.

The behavior of epistemic uncertainty is not static but varies significantly across domains and time scales, particularly in forecasting tasks involving chaotic systems. [64] demonstrates that in weather prediction, aleatoric uncertainty dominates on short timescales due to internal variability, whereas epistemic uncertainty becomes crucial on longer climate timescales where parameter constraints are weak. This temporal dynamic is supported by [16], which observes that while aleatoric uncertainty increases with the forecast horizon due to environmental variability, epistemic uncertainty remains relatively stable, reflecting the model's fixed knowledge state. However, contradictions exist regarding the reliability of these estimates under distributional shifts. [20] notes that common methods like Concrete Dropout and BNNs often fail to sufficiently increase uncertainty estimates when presented with test samples far from the training distribution, raising concerns about their safety in mission-critical applications. Furthermore, [287] challenges the assumption that epistemic uncertainty vanishes with infinite data in overparametrized networks, showing that non-identifiability due to symmetries can leave substantial residual parameter uncertainty even when the underlying function is fully identified.

Recent advancements also explore the integration of input uncertainty into the epistemic framework and the refinement of disentanglement metrics. [155] argues that propagating input uncertainty through a neural network inherently generates model uncertainty at the output, suggesting a unified formulation is necessary for stable decision boundaries. Additionally, [190] dissects epistemic uncertainty into procedural variability from training stochasticity and data variability, proposing influence function-based approaches to estimate these components directly. Despite these theoretical advances, the practical disentanglement of uncertainty types remains imperfect. [241] defines a specific metric for disentanglement error, revealing that even state-of-the-art Deep Ensembles exhibit room for improvement in

separating aleatoric and epistemic components. Consequently, while methods like [79] show promise by jointly training variance networks with BNNs, the field continues to grapple with ensuring that captured epistemic uncertainty accurately reflects the model's true state of knowledge rather than artifacts of approximation or architecture.

Challenges in Disentanglement and Advanced Frameworks

Despite the clear theoretical distinction between aleatoric uncertainty, which arises from inherent data noise, and epistemic uncertainty, stemming from model ignorance, practically disentangling these components remains a formidable obstacle in deep learning. A primary impediment is that standard metrics and approximation methods often fail to differentiate between these sources, exhibiting counter-intuitive behaviors that undermine their utility in safety-critical applications. Research has identified an "epistemic uncertainty hole" where epistemic metrics collapse in the presence of large models or limited training data, directly contradicting the theoretical expectation that model uncertainty should remain high when data is scarce [96]. This phenomenon is exacerbated by common approximation techniques; for instance, Monte Carlo Dropout frequently fails to capture increased uncertainty in extrapolation regions, rendering it unreliable for distinguishing true epistemic and aleatoric contributions compared to more rigorous Bayesian approaches [58]. Furthermore, reliance on information-theoretic measures for disentanglement has faced criticism due to the additivity assumption, which can render aleatoric and epistemic measures dependent rather than independent, leading to significant disentanglement errors even in state-of-the-art methods like Deep Ensembles [241].

The difficulty in separation is further compounded by the complex interplay between model architecture, dataset properties, and the resulting uncertainty landscapes. Studies indicate that prediction accuracy depends on both epistemic and aleatoric uncertainty in an intricate fashion that cannot be understood through marginalized uncertainty distributions alone; the correlation between accuracy and specific uncertainty types varies significantly across different neural network architectures and data distribution shifts [27]. Additionally, high model bias can lead to misleadingly low estimates of epistemic uncertainty, causing second-order uncertainty quantification methods to systematically blur bias-induced errors into aleatoric estimates, thereby underrepresenting the true scope of model ignorance [169]. In regression tasks specifically, restrictive assumptions regarding the likelihood function, such as Gaussianity, can result in improper capture of aleatoric uncertainty, which subsequently corrupts the estimation of epistemic uncertainty when the model is misspecified [111]. These findings suggest that without robust theoretical grounding and appropriate architectural choices, standard Bayesian Neural Networks (BNNs) and their approximations often provide a conflated view of predictive uncertainty. Moreover, standard calibration metrics often fail to account for model uncertainty, leading to incorrect conclusions about model reliability unless posterior predictive checks are employed [36].

To address these limitations, newer frameworks are emerging that utilize imprecise probability theory and advanced ensemble techniques to achieve a more rigorous separation of uncertainties. A notable advancement is Credal Bayesian Deep Learning (CBDL), which trains an uncountably infinite ensemble of BNNs using only finitely many elements via prior and likelihood finitely generated credal sets (FGCSs) [22]. By representing ambiguity in the selection of priors and likelihoods through convex combinations of extreme elements, CBDL generates a set of posteriors that allows for the distinct quantification of predictive aleatoric and epistemic uncertainties at inference time. This approach has been shown to be more robust to prior and likelihood misspecification and distribution shifts than single BNNs or standard ensembles, effectively overcoming the theoretical shortcomings of hierarchical Bayesian models while providing desirable probabilistic guarantees [22]. Similarly, cooperative training strategies have been proposed where a variance network is trained alongside a Bayesian neural network; this dual approach leverages the strengths of Mean Variance Estimation for aleatoric uncertainty and Bayesian inference for epistemic uncertainty, resulting in improved disentanglement and mean estimation across diverse datasets [79]. These methods address the failure of standard models to satisfy fundamental principles, such as the requirement that epistemic uncertainty must decrease with more data and increase with model complexity, often necessitating specialized regularization strategies like conflictual loss functions [69].

Beyond architectural innovations, gradient-based attribution methods are being developed to enhance the interpretability of uncertainty sources by identifying specific input regions contributing to each type of uncertainty. The proposed UABackprop method utilizes gradient-based uncertainty attribution to pinpoint problematic input regions, offering a mechanism to not only quantify but also explain and mitigate the effects of uncertainty on predictions [19]. This aligns with broader efforts to make uncertainty estimation actionable, allowing practitioners to understand whether uncertainty arises from data noise or model lack of knowledge in specific spatial or feature contexts. Furthermore, alternative frameworks such as Deep Evidential Regression attempt to learn evidential distributions for both uncertainty types directly, although recent analyses suggest these methods may function more as heuristics than exact quantification

tools due to gaps in their mathematical foundations [284]. Consequently, while methods like Deep Ensembles and Variational Inference remain prevalent, their ability to truly disentangle uncertainties is often contingent on specific regularization strategies and careful calibration [103]. The integration of these advanced frameworks and attribution techniques represents a critical shift toward more reliable, interpretable, and theoretically sound uncertainty quantification in deep learning systems, moving beyond simple predictive intervals to a nuanced understanding of the sources of error [86].

Applications in Science, Engineering, and Medicine

The deployment of Bayesian Deep Learning (BDL) and uncertainty quantification (UQ) in scientific and engineering domains has transitioned from theoretical exploration to a critical necessity for safety-critical decision-making. In fields where model errors can lead to catastrophic outcomes, such as structural health monitoring, medical diagnosis, and autonomous systems, the ability to distinguish between reducible epistemic uncertainty and irreducible aleatoric uncertainty is paramount. Traditional deep learning models often provide overconfident predictions, particularly when encountering out-of-distribution data, a flaw that Bayesian approaches aim to rectify by providing well-calibrated confidence estimates [8]. The practical utility of these methods is evident across diverse applications, ranging from computer vision in infrastructure inspection to complex physical simulations in nuclear engineering and climate science.

In the realm of computer vision and structural engineering, reliable uncertainty estimates are essential for automating inspection processes while maintaining safety standards. For instance, in structural health monitoring (SHM), deep vision models are employed to detect cracks and local damage in bridges and other critical infrastructure. Research demonstrates that incorporating Bayesian inference via Monte Carlo dropout allows these models to quantify prediction uncertainty, where metrics such as softmax variance and entropy correlate strongly with misclassifications [167]. This capability enables the system to trigger human intervention when uncertainty exceeds a specific threshold, thereby mitigating the risk of fatal oversight. Similarly, in robotics, uncertainty-aware controllers utilizing Control Barrier Functions (CBFs) rely on accurate variance estimates to guarantee safety. While methods like Gaussian Processes are computationally prohibitive for real-time control, and standard Monte Carlo dropout fails to capture aleatoric uncertainty, hybrid approaches combining deep ensembles and direct estimation have been shown to provide robust uncertainty estimates both in-domain and out-of-domain, significantly reducing failure rates in simulated robot operations [185]. Furthermore, in autonomous driving scenarios, advanced frameworks like Credal Bayesian Deep Learning have been applied to motion prediction, demonstrating superior robustness to distribution shifts and better disentanglement of uncertainty types compared to standard ensemble methods [22].

Medical imaging represents another domain where the stakes of uncertainty quantification are exceptionally high. The potential for deep learning models to "hallucinate" anatomical features during image reconstruction or denoising poses severe risks to patient safety. To address this, Bayesian approaches utilizing Deep Image Prior combined with Monte Carlo dropout have been developed to estimate both aleatoric and epistemic uncertainty in denoising tasks for X-ray, ultrasound, and OCT scans [199]. These methods ensure that the predictive uncertainty correlates with reconstruction errors, allowing clinicians to identify and disregard artifacts. In diagnostic applications, Bayesian models have been utilized to predict BI-RADS scores for mammographic masses, effectively mimicking the probabilistic reasoning of radiologists by reporting uncertainty levels associated with lesion malignancy [30]. Moreover, in stroke analysis, Bayesian Convolutional Neural Networks not only improve image-level accuracy but also provide patient-level uncertainty aggregation, enabling the filtering of critical diagnoses that require further expert examination [202]. The application of BDL extends to survival analysis as well, where models like NeuralSurv incorporate Bayesian principles to provide robust, well-calibrated uncertainty estimates for time-varying covariate-risk relationships, proving particularly valuable in data-scarce regimes [34].

In physics and environmental sciences, BDL serves as a bridge between data-driven efficiency and physical consistency. Deep neural networks are increasingly used as surrogate models for computationally expensive numerical simulations, such as those in inertial confinement fusion. By employing variational inference, these surrogates can generate calibrated Bayesian uncertainties that preserve physics-correlations, making comparisons between simulations and experiments meaningful [224]. In nuclear engineering, comparative studies of Monte Carlo dropout, deep ensembles, and Bayesian neural networks for predicting fission gas release and void fractions reveal that while all methods offer reasonable approximations, BNNs tend to produce larger, more conservative uncertainty estimates, which is preferable for safety assessments [78]. Similarly, in geosciences, Bayesian methods are applied to fault detection in seismic datasets and the estimation of essential climate variables from satellite observations. Here, the challenge lies in defining uncertainty in non-stationary Earth systems, where traditional distinctions between aleatoric and epistemic uncertainty may blur, necessitating a practical Bayesian perspective that interprets all uncertainty as rational belief informed by evidence [15]. Advanced frameworks also integrate symbolic scientific knowledge with

Bayesian deep learning to ensure physical constraints are respected, as seen in the Constraint-Aware Neurosymbolic Uncertainty Framework, which significantly reduces calibration errors in materials science and climate benchmarks [49].

Despite these successes, challenges remain regarding the reliability and calibration of different UQ methods. Empirical evaluations in radio galaxy classification and air quality forecasting indicate that while Bayesian neural networks often provide more reliable uncertainty estimates, scalable alternatives like deep ensembles and Monte Carlo dropout can perform competitively if applied correctly, though they may struggle with extrapolation regions or high-noise settings [228; 268]. Furthermore, studies on interatomic potentials in materials science highlight that the choice between variational BNNs and deep ensembles depends heavily on data representation and the specific requirements for out-of-distribution detection [5]. The ongoing development of libraries like Fortuna and the refinement of techniques such as density uncertainty layers continue to enhance the robustness of these models, ensuring that uncertainty estimates are grounded in the empirical density of the input data [73; 177]. As the field progresses, the focus is shifting towards active learning frameworks where uncertainty guides data annotation, and towards hybrid models that combine the scalability of deep learning with the rigorous probabilistic foundations of Bayesian inference to handle the complexities of real-world scientific data [146].

Computer Vision, Autonomous Systems, and Robotics

In the deployment of deep learning within safety-critical robotic systems and autonomous vehicles, the ability to quantify uncertainty is a prerequisite for reliable operation rather than a mere theoretical enhancement. Unlike traditional software, deep neural networks often produce overconfident predictions when encountering out-of-distribution data or ambiguous sensory inputs, posing significant risks in applications ranging from autonomous driving to medical robotics [8; 3]. The fundamental challenge lies in distinguishing between aleatoric uncertainty, which arises from inherent noise in the data such as sensor limitations or conflicting labels, and epistemic uncertainty, which stems from model limitations and can be reduced with more diverse training data [8; 290]. Effective uncertainty quantification enables autonomous agents to recognize their own limitations, facilitating safer decision-making processes such as yielding control to human operators or rejecting unreliable sensor data [6; 93].

Affordance prediction, a critical capability for robots to understand how to interact with objects, has traditionally relied on deterministic models that fail to capture the stochastic nature of real-world environments. Recent advancements have introduced Bayesian deep learning methods to predict affordances directly from RGB images, explicitly modeling both aleatoric and epistemic uncertainties to ensure robust interaction [6]. Comparative studies in this domain indicate that while both Monte Carlo dropout and deep ensembles are effective, deep ensembles often yield marginally better performance in terms of calibration and Brier scores, suggesting a superior ability to capture the complexity of socially acceptable behaviors and object interactions [6]. This aligns with broader findings in computer vision where deep ensembles are frequently regarded as the state-of-the-art for uncertainty quantification, despite their higher computational cost compared to single-model approximations like Monte Carlo dropout [76; 48].

However, the reliability of these methods is not uniform across all tasks, particularly in dense stereo matching and depth estimation where standard approaches often neglect epistemic uncertainty. This omission is particularly detrimental when test domains differ from training distributions, such as in colonoscopy depth estimation where lighting and texture conditions vary drastically [125; 187]. Research demonstrates that incorporating epistemic uncertainty via probabilistic deep learning or deep ensembles significantly improves the correspondence between estimated uncertainty and actual error rates, especially in regions with repetitive patterns or lack of texture [125]. Furthermore, in medical imaging tasks like General Movement Assessment, disentangling these uncertainty sources allows for better class separation and more reliable clinical decisions, highlighting the necessity of tailored uncertainty quantification strategies for specific modalities [263].

The application of uncertainty quantification extends critically to control systems, where safety is paramount. In robotics, Control Barrier Functions rely on accurate uncertainty estimates to guarantee safe operation. Traditional methods using Gaussian Processes are computationally prohibitive for high-dimensional sensory data, while Monte Carlo dropout fails to capture aleatoric uncertainty. Hybrid approaches combining model variance-based algorithms with direct estimation techniques have shown to provide more accurate in-domain and out-of-domain uncertainty, resulting in safer robot controllers with lower failure rates [185]. Similarly, in trajectory prediction for maritime surveillance, recurrent encoder-decoder networks enhanced with Bayesian modeling of both uncertainty types improve prediction reliability, particularly when additional intention data is available [149].

Nevertheless, contradictions and challenges persist regarding the evaluation and calibration of these uncertainty estimates. Metrics such as Calibration Error are widely used but can be misleading if model uncertainty is not properly accounted for, potentially leading to trust in flawed models [36]. Studies comparing various uncertainty quantification methods on regression tasks reveal that no single method universally outperforms others; while deep ensembles and Monte Carlo dropout tend to be more conservative, heteroscedastic neural networks can achieve strong results with less complexity, though they may struggle to separate uncertainty sources effectively [48; 171]. Furthermore, the assumption that increasing the number of ensemble members always improves uncertainty estimation has been challenged, with evidence suggesting that without post-hoc calibration, larger ensembles may underestimate uncertainty, leading to unintended exploratory behavior in optimization tasks [159].

Future directions in this field emphasize the need for scalable, efficient, and theoretically grounded methods that can operate in real-time on embedded systems. Innovations such as hyper-diffusion models aim to estimate both uncertainty types with a single model, bypassing the computational burden of large ensembles while maintaining accuracy

comparable to multi-model approaches [201]. Additionally, the development of standardized benchmarks and toolkits is crucial for reproducibility and fair comparison across different architectures and application domains, ensuring that uncertainty quantification methods are rigorously validated before deployment in high-stakes scientific and engineering contexts [67; 12]. As autonomous systems become more pervasive, the integration of robust uncertainty quantification will remain a cornerstone for achieving trustworthy and safe artificial intelligence.

Medical Imaging, Healthcare, and Diagnostics

The deployment of artificial intelligence in clinical settings necessitates a fundamental shift from deterministic point estimates to probabilistic predictions that explicitly quantify reliability. In high-stakes domains such as medical imaging and diagnostics, the distinction between aleatoric uncertainty, which arises from inherent data noise or anatomical ambiguity, and epistemic uncertainty, stemming from model ignorance or limited training data, is not merely a theoretical construct but a prerequisite for safe decision support [8; 86]. Traditional deep learning models frequently exhibit overconfidence, particularly when encountering out-of-distribution samples or heterogeneous patient data, posing significant risks in automated diagnosis. Consequently, Bayesian methods and ensemble techniques have become essential for propagating uncertainty through diagnostic pipelines, enabling systems to identify ambiguous cases that require expert review and optimizing the allocation of clinical resources [148; 9].

In the realm of image segmentation, which serves as the foundation for extracting quantitative imaging biomarkers, uncertainty estimation functions as a critical quality control mechanism. Segmentation errors can propagate significantly into downstream statistical analyses, leading to unfaithful biomarker estimates that may misguide treatment plans. Research demonstrates that Bayesian neural networks (BNNs) can sample from the posterior distribution of segmentations to generate voxel-wise uncertainty maps, which are then aggregated into organ-level confidence measures [252]. By integrating these confidence measures into group analyses and disease classification models, studies have shown improved alignment with manual references and higher classification accuracy compared to deterministic approaches. Various architectures, including Monte Carlo Dropout, Probabilistic U-Nets, and Hierarchical Probabilistic U-Nets, have been evaluated to capture multi-scale ambiguities inherent in medical images, such as liver segmentation in diabetic patients [252; 9]. Furthermore, recent advancements allow for the efficient estimation of uncertainty in state-of-the-art frameworks like nnU-Net without altering their original architecture, utilizing trajectory-based posterior sampling during stochastic gradient descent to preserve performance while adding robustness [244]. This approach addresses the limitations of standard variational inference, which can be unstable, and Monte Carlo Dropout, which has been found to fail in capturing true uncertainty in extrapolation regions [58; 244].

Diagnostic applications extend beyond segmentation to include disease detection and triage, where the distinction between uncertainty types directly informs clinical workflow. In stroke detection using MRI, Bayesian convolutional neural networks provide probability maps of lesions alongside uncertainty estimates, enabling the aggregation of image-level predictions to the patient level [202]. This aggregation effectively filters critical diagnoses that exhibit high uncertainty for further examination by physicians, thereby reducing the risk of false negatives in emergency settings. Similarly, in breast cancer screening, uncertainty-aware models have been developed to predict BI-RADS scores, mimicking the probabilistic reasoning of radiologists. These models utilize morphological features to report uncertainty regarding malignancy, achieving performance that rivals or exceeds human experts in specific categories while providing a transparent measure of confidence [30; 204]. The ability to flag uncertain cases is particularly valuable in active learning setups, where the model queries the most ambiguous samples for labeling, thereby optimizing the use of expert annotation resources in data-scarce medical domains [146; 270].

Survival analysis and prognosis modeling also benefit significantly from uncertainty quantification, particularly in chronic disease management. In Alzheimer's disease progression modeling, comparative studies of Monte Carlo Dropout, Variational Inference, and Markov Chain Monte Carlo reveal that probabilistic approaches produce well-calibrated confidence bounds for cognitive score trajectories, even under noisy training conditions [Uncertainty Quantification in Alzheimer's Disease Progression Modeling]. Bayesian frameworks for survival models, combining variational inference with neural multi-task logistic regression, offer not only accurate survival predictions but also a quantification of survival uncertainty, which is crucial for developing personalized treatment protocols [56; 215]. Additionally, in drug discovery, probabilistic predictive models help mitigate the risks of over-confidence by returning distributions of predicted values, allowing researchers to account for uncertainty when making expensive investment decisions regarding compound development [211].

Despite these advances, significant challenges remain in the calibration and interpretation of uncertainty metrics, particularly regarding their alignment with human judgment. Studies indicate that standard calibration errors may be misleading when model uncertainty is present, necessitating the use of posterior predictive checks to avoid mistrusting good models or trusting bad ones [36]. Furthermore, the relationship between predictive uncertainty and human

linguistic uncertainty in radiology reports shows only modest correlation, highlighting the difficulty in aligning machine-generated confidence with human interpretative nuances [197]. There is also ongoing debate regarding the effectiveness of different approximation methods; while Monte Carlo Dropout is computationally efficient, findings suggest it may fail to capture true uncertainty compared to more rigorous Bayesian approaches or deep ensembles, which can be viewed as Gaussian Process approximate posteriors [58; 239]. Nevertheless, the consensus emphasizes that distinguishing between data noise and model ignorance allows for more robust decision support systems, enabling the rejection of unreliable predictions and the detection of out-of-distribution samples in safety-critical medical applications [103; 135].

Materials Science, Physics, and Environmental Modeling

In the domains of materials science and computational physics, the deployment of machine learning interatomic potentials (MLIPs) has transformed the ability to simulate atomic energies and forces with near-quantum accuracy, yet the utility of these surrogate models hinges on their capacity to quantify predictive uncertainty, particularly when extrapolating to out-of-distribution structures common in materials discovery. Systematic comparisons between variational Bayesian neural networks and deep ensembles reveal that while both approaches offer pathways to uncertainty estimation, their performance is highly sensitive to data representation and the specific inference algorithm employed. For instance, in predicting stress fields for diverse material microstructures, rigorous inference methods such as Hamiltonian Monte Carlo and Bayes by Backprop have been shown to provide consistent uncertainty estimates, whereas Monte Carlo Dropout often yields results that are difficult to interpret and heavily dependent on design choices [145]. This inconsistency is further corroborated in nuclear engineering applications, where Bayesian neural networks generally produce larger, more conservative uncertainties compared to Dropout and ensemble methods when predicting neutron flux profiles, highlighting the critical sensitivity of safety-critical predictions to the chosen inference technique [91]. Furthermore, the integration of physics constraints into these models is gaining traction; frameworks like CANUF unify Bayesian deep learning with differentiable symbolic reasoning to ensure physical consistency, significantly reducing calibration errors in materials benchmarks while respecting domain constraints [49]. Similarly, Bayesian Physics-Informed Neural Networks (B-PINNs) have demonstrated superior reliability in prognostics for transformer ageing by embedding heat diffusion partial differential equations, accurately quantifying predictive uncertainty where dropout-based baselines fail [74].

The stakes for reliable uncertainty quantification are equally high in environmental modeling and geosciences, where the chaotic nature of atmospheric systems demands probabilistic approaches that traditional deterministic deep learning cannot provide. Deep learning is increasingly utilized to estimate Essential Climate Variables (ECVs) from satellite observations, yet the quantification of uncertainties associated with these outputs has historically lagged behind improvements in accuracy, necessitating a bridge between conventional statistical views and modern deep learning techniques [24]. Recent advancements have successfully bridged the gap between Bayesian deep learning and traditional ensemble weather forecasting by explicitly decomposing predictive uncertainty into epistemic and aleatoric components. By utilizing variational inference for model uncertainty and physics-informed stochastic perturbations for data noise, these hybrid frameworks achieve superior computational efficiency and calibration compared to standard probabilistic diffusion models [63]. This decomposition is vital for subseasonal forecasting, where calibrated confidence in temperature anomaly predictions is a prerequisite for detecting extreme heat events, necessitating architectures like Bayesian UNet++ that balance prediction error with calibration reliability [264]. Similarly, in hydrological modeling, deep learning baselines such as Mixture Density Networks and Monte Carlo Dropout have been benchmarked to demonstrate that accurate and precise uncertainty estimation for rainfall-runoff modeling is achievable, addressing a long-standing gap in community standards for hydrological uncertainty [272].

The extension of these methods to natural hazard forecasting further underscores the necessity of disentangling uncertainty types for effective decision support. In wildfire danger forecasting, uncertainty-aware frameworks that jointly capture epistemic and aleatoric uncertainties have demonstrated improved predictive skill and calibration, revealing that while aleatoric uncertainty increases with the forecast horizon due to environmental variability, epistemic uncertainty remains stable, providing complementary insights for risk management [16]. However, contradictions persist regarding the efficacy of approximate inference methods across these domains. While deep ensembles are often considered state-of-the-art, theoretical analyses suggest they can be viewed as approximate Bayesian methods, and refinements incorporating Gaussian process approximations can yield more reliable uncertainties than standard implementations [239]. Conversely, empirical studies indicate that Monte Carlo Dropout struggles to accurately reflect true uncertainty, particularly failing to capture increased uncertainty in extrapolation regions compared to traditional Bayesian approaches, a limitation observed in both general regression tasks and specific geophysical applications [58]. To address the computational barriers of rigorous Bayesian inference in these large-scale simulations, scalable alternatives such as implicit Bayesian neural networks via layer-wise input augmentation have been developed, achieving state-of-the-art calibration without the impractical inference over millions of parameters [174]. Ultimately, the synthesis of these diverse applications underscores the necessity of selecting uncertainty quantification strategies

that are not only computationally feasible but also theoretically grounded in the specific physical constraints and data characteristics of the problem at hand, whether modeling atomic interactions or global climate dynamics.

Industrial, Business, and Financial Applications

The adoption of Bayesian Deep Learning (BDL) in industrial and financial sectors is driven by the imperative to transform predictive models from opaque forecasting tools into robust decision-support systems capable of quantifying risk. In these high-stakes environments, the distinction between aleatoric (data) and epistemic (model) uncertainty is not merely theoretical but operational, directly influencing resource allocation and risk mitigation strategies. A primary application of this paradigm is found in business process monitoring, where organizations aim to predict the remaining execution time of ongoing cases. Traditional neural networks often provide point estimates without confidence intervals, leading to costly errors when models encounter unfamiliar data patterns. Research demonstrates that applying Bayesian neural networks and concrete dropout to predictive process monitoring allows for the estimation of both uncertainty types, enabling the separation of high-quality predictions from low-quality ones [55]. This capability facilitates strategic interventions, such as rerouting cases with long expected remaining times or high uncertainty to specialized tracks, thereby optimizing operational efficiency and fostering better human-AI cooperation [33].

In the domain of customer relationship management, accurate prediction of Customer Lifetime Value (LTV) is essential for revenue optimization. Standard deep learning models frequently fail to capture the inherent uncertainty in user behavior, resulting in overconfident predictions that can misguide marketing spend. By incorporating the Monte Carlo Dropout (MCD) framework, recent approaches have enhanced LTV prediction architectures to provide confidence metrics alongside point estimates. This advancement allows businesses to target high-value customers with greater assurance and avoid investing in segments where the model exhibits high uncertainty, ultimately improving the Top 5% Mean Absolute Percentage Error compared to state-of-the-art deterministic methods [53]. However, the reliability of such uncertainty estimates is paramount; empirical comparisons suggest that while MCD is computationally efficient, it may struggle to accurately reflect increased uncertainty in extrapolation regions compared to more rigorous Bayesian approaches, posing risks in financial scenarios involving novel market conditions [58].

Similarly, in the financial sector, particularly for Small and Medium Enterprises (SMEs), revenue forecasting is a key indicator for credit limit management. The development of scalable natural gradient boosting machines that yield high-quality predictive uncertainty estimates enables banks to distinguish between accurate and inaccurate forecasts, ensuring that credit limits are granted based on reliable risk assessments rather than potentially flawed point predictions [254]. The choice of uncertainty measure is critical in these safety-critical applications; research utilizing proper scoring rules and kernel scores provides a unified framework for designing measures that balance robustness, tail sensitivity, and out-of-distribution responsiveness, which is vital for preventing catastrophic losses in financial ecosystems [279].

The energy and utility sectors also benefit significantly from uncertainty-aware forecasting, particularly in electricity price prediction and grid management. Probabilistic forecasts are fundamental for energy risk management, yet many models fail to capture the full extent of uncertainty arising from data noise and model choices. Studies comparing statistical and deep learning models, such as deep distributional neural networks augmented with ensemble approaches or conformal prediction, reveal that accounting for both data and model uncertainty improves both point and probabilistic forecasting accuracy [298]. Furthermore, in low-voltage state estimation for distribution system operators, the activation of flexible resources introduces significant volatility. Bayesian neural networks have been shown to capture flexibility-induced voltage drops by dynamically extending prediction intervals during activation periods, thereby providing the interpretability and reliability required for robust grid decision-making [184].

Moreover, the application of uncertainty quantification extends to predictive maintenance and asset health monitoring, where estimating the Remaining Useful Life (RUL) of machinery is paramount. In these contexts, the ability to decompose total predictive uncertainty into aleatoric and epistemic components allows for more informed maintenance scheduling. For instance, in battery health monitoring, Bayesian neural networks that output distributions rather than single points have demonstrated the ability to capture inherent randomness in battery degradation, significantly improving prediction certainty from the initial to the mid-life stages of the battery [84]. Similarly, in aircraft engine prognostics, the separation of uncertainty types aids in active learning strategies, where radial Bayesian neural networks and Monte Carlo Dropout are identified as effective options for identifying data points that would most reduce model uncertainty, though no single method consistently outperforms others across all situations [48]. The convergence of these methodologies underscores a broader trend where uncertainty quantification is not merely an academic exercise but a foundational requirement for deploying trustworthy AI in industrial, business, and financial ecosystems, enabling

strategies that account for prediction risk and prevent catastrophic losses by signaling when models are operating outside their validity range.

Evaluation Metrics, Calibration, and Benchmarking

The rigorous evaluation of uncertainty quantification (UQ) methods remains a foundational challenge in deploying reliable machine learning systems, particularly because standard accuracy metrics often fail to capture the reliability of predictive distributions in high-stakes environments. A primary concern in this domain is calibration, which assesses the alignment between predicted confidence and actual empirical accuracy. While metrics such as Expected Calibration Error (ECE) are widely adopted for classification, their application to regression and their behavior in the presence of model uncertainty are subjects of intense scrutiny. Research indicates that calibration error can be misleading when epistemic uncertainty is significant, as it often conflates reducible model uncertainty with irreducible aleatoric noise, potentially leading to misplaced trust in models that appear well-calibrated but fail to recognize their own ignorance [36]. Consequently, there is a growing consensus that evaluation protocols must distinguish between these uncertainty sources to avoid penalizing models that correctly express high uncertainty on out-of-distribution data. Furthermore, recent work distinguishes between Uncertainty Quantification (UQ) and Robustness Quantification (RQ), demonstrating that while UQ assesses predictive confidence, RQ evaluates sensitivity to input perturbations, and that a combination of both approaches often yields superior reliability assessments compared to either method alone [243].

To address the limitations of single-metric evaluations, the literature proposes a diverse array of specialized metrics tailored to specific task requirements. In regression settings, metrics such as the Area Under Sparsification Error (AUSE), Negative Log-Likelihood (NLL), and Calibration Error (CE) are frequently employed to assess the quality of predictive intervals. Comparative studies suggest that while CE offers stability and interpretability, AUSE provides a more robust measure for evaluating how well uncertainty estimates correlate with prediction errors, leading some researchers to recommend replacing Spearman's Rank Correlation with AUSE for regression tasks [171]. Furthermore, the framework of proper scoring rules, including the Continuous Ranked Probability Score (CRPS) and logarithmic scores, offers a unified theoretical basis for evaluating both accuracy and reliability simultaneously. These scoring rules enable the decomposition of total uncertainty into aleatoric and epistemic components, providing a principled approach to benchmarking regression models that goes beyond simple interval coverage [164; 279]. However, the theoretical underpinnings of common measures like entropy and variance have been challenged; axiomatic assessments reveal that many widely used metrics lack formal justification in regression settings and may behave unpredictably under distribution shifts, necessitating a shift toward measures grounded in predictive exponential families [207].

Despite the availability of these metrics, significant contradictions exist regarding their effectiveness across different architectures and data regimes. The difficulty of disentangling uncertainties is further highlighted by findings that information-theoretic measures often rely on additivity assumptions that do not hold in practice, prompting the development of new metrics like Disentanglement Error to specifically benchmark the quality of uncertainty separation [241]. Additionally, information-theoretic frameworks suggest that the choice of uncertainty measure must align with the predicting model, as the disentanglement between aleatoric and epistemic components varies substantially between in-distribution and out-of-distribution data [80]. Critical analysis reveals that current methods often fail to capture critical components of epistemic uncertainty, particularly model bias, which is systematically blurred into aleatoric estimates, thereby underrepresenting the true epistemic risk [169]. For instance, while Deep Ensembles are often considered the state-of-the-art for UQ, empirical evaluations reveal that increasing the number of ensemble members can paradoxically lead to an underestimation of uncertainty if post-hoc calibration is not applied [159].

The challenge of evaluation is compounded in domains where ground truth is ambiguous or non-existent, such as medical imaging and climate science. In these fields, the lack of clear labels necessitates the use of proxy metrics or synthetic benchmarks where noise levels can be controlled. Studies in medical imaging demonstrate that even when models achieve high accuracy, their uncertainty estimates may be systematically underestimated, requiring simple scaling techniques like σ -scaling to recalibrate predictive distributions for reliable decision support [103]. Moreover, the correlation between machine-derived uncertainty and human linguistic uncertainty in clinical reports is often modest, suggesting that current metrics may not fully capture the nuances of human expert hesitation [197]. In earth observation, the dynamic nature of the data requires benchmarks that account for spatiotemporal shifts, yet standard UQ methods often fail to adapt to the specific noise characteristics of satellite-derived essential climate variables [24]. In such contexts, metrological frameworks are being proposed to extend concepts of uncertainty

evaluation to nominal properties, enabling rigorous assessment in classification tasks where traditional error bars are undefined [192].

To facilitate fair comparisons and reproducibility, the community has begun developing standardized toolkits and benchmarking frameworks. Surveys of available software highlight a fragmentation in the ecosystem, with tools like Pyro and TensorFlow Probability offering flexibility, while others like Uncertainty Quantification 360 provide broader methodological scope [67]. Recent efforts propose unified frameworks that generate predictive uncertainty measures via Bayesian estimation, allowing for consistent evaluation across different downstream tasks such as out-of-distribution detection and misclassification identification [193]. Libraries such as Fortuna have emerged to support scalable Bayesian inference and calibration techniques like conformal prediction, simplifying the process of benchmarking for practitioners [73]. The establishment of such axiomatic foundations and standardized protocols is critical for advancing the field, as it allows researchers to move beyond ad-hoc evaluations toward a more comprehensive understanding of when and why specific UQ methods succeed or fail. Ultimately, the path forward requires not only the refinement of existing metrics but also the creation of benchmarks that explicitly test for robustness against procedural variability, data bias, and the complex interplay between model architecture and uncertainty sources [169; 190].

Calibration and Sharpness Metrics

The evaluation of uncertainty quantification in deep learning relies fundamentally on the dual concepts of calibration and sharpness, yet the interplay between these metrics reveals significant complexities when distinguishing between aleatoric and epistemic uncertainty. Calibration metrics, such as Expected Calibration Error (ECE) and Negative Log-Likelihood (NLL), are traditionally employed to assess whether a model's predicted confidence aligns with its actual accuracy. While some empirical studies suggest that Calibration Error is the most stable and interpretable metric for regression tasks, offering robustness across varying test set sizes [171], this view is increasingly challenged by theoretical analyses. Critics argue that standard calibration errors can be fundamentally misleading in the presence of significant model uncertainty, as they often conflate irreducible data noise (aleatoric) with model ignorance (epistemic) [36]. This conflation leads to a paradoxical situation where a model that correctly reflects high inherent ambiguity in the data may be penalized as poorly calibrated, while an overconfident model that ignores uncertainty might appear deceptively reliable if the evaluation metric assumes a deterministic ground truth. Consequently, relying solely on scalar calibration metrics can result in incorrect conclusions regarding model trustworthiness, particularly in safety-critical domains where distinguishing between what the model knows and what it does not is paramount.

The concept of sharpness, defined as the concentration of the predictive distribution, introduces a critical trade-off that complicates the pursuit of perfect calibration. A model that is perfectly calibrated but unsharp provides limited utility, effectively predicting the marginal distribution of the data without leveraging input-specific information. Conversely, an overly sharp model risks being overconfident and inaccurate, particularly on out-of-distribution samples. Recent literature emphasizes that achieving reliable uncertainty quantification requires balancing these two properties, often necessitating a slight compromise in prediction error to achieve sharper, more reliable forecasts with high coverage probabilities [264]. In medical imaging and other safety-critical applications, methods that optimize for sharpness without proper regularization can lead to underestimated variances, whereas those focusing solely on calibration may produce overly conservative prediction intervals that fail to inform decision-making [103]. This tension is further exacerbated by findings that high model bias can lead to misleadingly low estimates of epistemic uncertainty, causing second-order methods to blur bias-induced errors into aleatoric estimates, thereby underrepresenting the true scope of model ignorance [169].

To address the limitations of scalar metrics like ECE and NLL, recent work has proposed moving toward unified frameworks based on proper scoring rules and kernel scores. These approaches offer a more principled method for evaluating both calibration and sharpness simultaneously while facilitating the decomposition of total uncertainty into aleatoric and epistemic components. Specifically, the use of kernel scores allows for the design of uncertainty measures where characteristics such as tail sensitivity, robustness, and out-of-distribution responsiveness are governed by the choice of the kernel function [279]. This framework unifies several well-known measures and provides a principled recipe for designing task-specific metrics that can distinguish whether a calibration failure stems from data noise or model inadequacy [164]. By deriving closed-form expressions for uncertainty measures under practical parametric assumptions, these methods recover popular regression UQ measures based on predictive variance and differential entropy while ensuring they satisfy rigorous axiomatic properties [207]. Furthermore, proper scoring rules such as the Continuous Rank Probability Score (CRPS) have been utilized to derive cost functions that explicitly encode the trade-off between accuracy and reliability, enabling the estimation of input-dependent variance in deterministic models where ensemble methods are computationally prohibitive [162].

Despite these advancements, contradictions remain regarding the efficacy of specific metrics across different domains and the behavior of epistemic uncertainty measures. While some studies advocate for the stability of Calibration Error, others demonstrate that common measures of epistemic uncertainty, such as those derived from Deep Ensembles, can violate fundamental principles—such as decreasing uncertainty with more data or increasing uncertainty with model complexity—suggesting that current calibration techniques may not adequately address underlying posterior approximation errors [69]. The relationship between sharpness and calibration is further complicated by the finding that standard calibration techniques often fail to account for the irreducible noise present in the data-generating process, leading to metrics that penalize models for correctly reflecting data ambiguity. Additionally, the challenge of disentangling uncertainty sources is critical; frameworks that decompose statistical pointwise risk into aleatoric and epistemic components via Bayesian estimation have shown promise in detecting out-of-distribution instances, yet the

alignment of these measures with the predicting model varies substantially between in-distribution and out-of-distribution data [193; 80]. Consequently, the development of metrics that can disentangle the effects of aleatoric and epistemic uncertainty while rewarding both calibration and sharpness is paramount. Future evaluation protocols must move beyond simple scalar summaries to adopt standardized suites that assess the robustness of uncertainty estimates under distributional shifts, ensuring that models deployed in high-stakes environments are trustworthy not just in their average performance, but in their ability to recognize and quantify their own limitations [48; 130].

Benchmarking Frameworks and Pitfalls

The systematic evaluation of Uncertainty Quantification (UQ) methods in deep learning has evolved through the development of specialized frameworks designed to assess both synthetic and real-world performance, yet significant challenges remain in establishing a universal standard. To address the fundamental difficulty of lacking ground-truth uncertainty in many practical applications, researchers have proposed frameworks based on synthetic regression problems where the underlying functions and noise distributions are explicitly known. These synthetic environments are crucial because they enable the precise decomposition of total uncertainty into its constituent aleatoric and epistemic components, a task that is often impossible with standard real-world datasets where only the aggregate error is observable [12]. By utilizing regression problems where the function is a linear combination of nonlinear functions, these frameworks allow for the comparison of deep learning UQ results against statistical reference methods based on Bayesian linear regression, providing a reliable baseline for assessing coverage probabilities and uncertainty size [12]. Complementing these synthetic approaches, other frameworks utilize real-world datasets augmented with crowd-sourced labels or multiple annotators to establish a proxy for intrinsic data ambiguity. For instance, studies comparing Deep Ensembles and Deep Evidential Regression have leveraged injected noise levels in both zero-dimensional and two-dimensional data to evaluate how well predicted aleatoric uncertainty scales with known ground-truth noise, revealing that while uncertainty often scales correctly, it frequently remains miscalibrated in high-noise and high-dimensional settings [120]. Despite these methodological advances, the field currently suffers from a fragmentation of benchmarks, leading to inconsistent comparisons where methods are evaluated on disparate tasks with incompatible metrics, making it difficult to select appropriate UQ methods for specific applications [8].

A significant pitfall pervading current evaluation practices is the reliance on metrics that implicitly assume a single correct label or a specific parametric form for the predictive distribution, which fails to capture tasks characterized by multiple valid outcomes or high intrinsic noise. Standard metrics like Mean Squared Error (MSE) or Negative Log-Likelihood (NLL) often presuppose Gaussianity, which can lead to misleading conclusions when the true data distribution is multimodal, heavy-tailed, or subject to model misspecification [207]. This limitation is particularly acute in regression tasks where the assumption of homoscedastic noise is frequently violated; many benchmarking protocols do not adequately penalize methods that fail to model heteroscedasticity correctly, leading to unreliable epistemic uncertainty estimates when the underlying model assumptions are incorrect [111]. To address these deficiencies, the use of proper scoring rules, such as the Continuous Ranked Probability Score (CRPS) and various kernel scores, has been advocated to provide a more rigorous assessment that does not rely on strict distributional assumptions. These unified frameworks based on proper scoring rules allow for the derivation of closed-form expressions for uncertainty measures that naturally decompose into aleatoric and epistemic components, offering a principled recipe for designing new measures with specific behaviors regarding tail sensitivity and robustness [164; 279]. However, the adoption of these superior metrics in standard benchmarks remains inconsistent, perpetuating the use of less informative evaluation criteria.

Another critical failure mode identified in benchmarking studies is the inability of many UQ methods to appropriately increase uncertainty estimates for out-of-distribution (OOD) samples. While theoretical frameworks suggest that epistemic uncertainty should rise as inputs deviate from the training manifold, empirical evaluations frequently reveal that popular methods, including Monte Carlo Dropout and certain Bayesian Neural Network approximations, remain overconfident on OOD data [20]. This phenomenon is exacerbated by the fact that some methods conflate model bias with aleatoric uncertainty, leading to systematically underestimated epistemic uncertainty even when the model is clearly operating outside its domain of competence [169]. The discrepancy between theoretical expectations and empirical behavior highlights a gap in current benchmarking suites, which often prioritize in-distribution calibration metrics over robustness to distributional shifts [130]. Furthermore, the lack of universal benchmarks has led to difficulties in comparing the computational efficiency and scalability of different UQ approaches. While some methods offer superior uncertainty calibration, they may be computationally prohibitive for large-scale deployment, a trade-off that is not always captured in standard accuracy-focused evaluations [67]. For example, deep ensembles often provide robust uncertainty estimates but at a high computational cost, whereas single-model approaches like Deep Evidential Regression offer efficiency but may rely on heuristic assumptions that break down under rigorous stress testing [284].

Recent efforts to introduce axiomatic assessments aim to resolve these inconsistencies by defining a set of desirable properties that any valid uncertainty measure must satisfy, such as monotonicity with respect to noise levels and sensitivity to model capacity [207]. However, the absence of a standardized protocol for applying these axioms across diverse architectures and domains continues to hinder the development of universally reliable UQ methods. The evaluation of UQ methods is further complicated by the interplay between different sources of uncertainty, where errors in estimating one component can mask deficiencies in the other. Studies have shown that methods claiming to disentangle aleatoric and epistemic uncertainty often fail to do so in practice, particularly when the model architecture introduces significant bias that is misattributed to data noise [169]. This misattribution can lead to false confidence in safety-critical applications, where distinguishing between reducible model uncertainty and irreducible data noise is paramount for decision-making [86]. Consequently, there is a growing consensus that future benchmarking frameworks must incorporate more granular evaluation metrics that can separately assess the fidelity of aleatoric and epistemic estimates, rather than relying solely on aggregate measures of predictive performance [48]. Without such refined tools, the field risks stagnating with methods that appear robust on standard benchmarks but fail to provide reliable uncertainty estimates in complex, real-world scenarios.

Comparative Studies of UQ Methods

Comparative analyses across diverse scientific and engineering domains consistently demonstrate that no single uncertainty quantification (UQ) method universally dominates; rather, performance is highly contingent on the specific application, data characteristics, and the nature of the uncertainty being modeled. In the realm of computer vision and complex physical simulations, deep ensembles frequently outperform variational methods, particularly in out-of-distribution (OOD) detection tasks. For instance, evaluations on complex particle classification tasks using liquid argon time projection chamber data reveal that ensembling methods achieve better calibration and generally superior performance compared to other existing techniques [147]. This superiority is further corroborated in hyperspectral image target detection, where bootstrapped neural networks and deep ensembles closely rival the gold standard of Markov Chain Monte Carlo (MCMC) while offering significantly reduced computational expenses [161]. Theoretical work suggests that deep ensembles implicitly perform empirical Bayes procedures with data-dependent priors, providing a principled explanation for their robust empirical performance compared to standard Bayesian neural networks (BNNs) [182]. However, this dominance is not absolute; in the context of machine learning interatomic potentials, while deep ensembles and variational BNNs show comparable performance, the choice often depends on specific requirements for extrapolation and data representation, with BNNs sometimes producing larger uncertainties than ensembles [5].

Conversely, Monte Carlo (MC) dropout remains a popular choice due to its simplicity, offering a pragmatic balance between efficiency and performance for certain vision and medical imaging tasks, though its reliability is not uniform. Studies in quantitative MRI indicate that MC dropout can significantly improve accuracy while providing effective uncertainty maps that highlight pathological regions, even with limited training data [283]. Similarly, in nuclear engineering applications involving fission gas release, MC dropout and BNNs produce uncertainty bands that agree well and envelop noisy measurement data, suggesting reasonable estimation capabilities [78]. However, qualitative and empirical analyses suggest that MC dropout can struggle to accurately reflect true uncertainty, particularly failing to capture increased uncertainty in extrapolation regions where Bayesian models succeed [58]. Furthermore, the quality of uncertainty estimated via MC dropout is sensitive to architectural choices and training procedures, necessitating careful tuning to avoid unsatisfactory results [152]. To address efficiency concerns in scientific computing, recent advancements such as Multi-Level Monte Carlo dropout have been proposed to reduce the computational cost of estimating predictive moments by reusing dropout masks across fidelity levels, demonstrating significant efficiency gains over single-level approaches [274].

In physical systems and scientific computing, the sensitivity of UQ performance to data characteristics becomes particularly pronounced, highlighting critical pitfalls in method selection. Comparative studies involving simple physical systems, such as a single pendulum, reveal that when training data exhibits low variance in noise, methods like BNNs, Concrete Dropout, and Deep Ensembles may fail to capture input-dependent noise, predicting constant relative uncertainty regardless of the input [20]. This issue is particularly hard to avoid in BNNs, whereas Concrete Dropout struggles significantly when test samples lie far from the training distribution. In the context of rainfall-runoff modeling, while MC dropout and Mixture Density Networks can achieve reliable uncertainty estimation, there is a noted tendency for some methods to underestimate uncertainties if aleatoric components are not explicitly modeled [272]. Research into aleatoric uncertainty estimation reveals that while methods like Deep Ensembles and Deep Evidential Regression can scale predicted uncertainty with injected noise levels, they often suffer from miscalibration, particularly in high-noise and high-dimensional settings [120]. In medical imaging regression tasks, predictive uncertainty is systematically underestimated by variational Bayesian inference with MC dropout, requiring post-hoc calibration techniques like sigma scaling to ensure robust rejection of unreliable predictions [103].

The distinction between aleatoric and epistemic uncertainty further complicates method selection, especially in prognostics and active learning. For tasks requiring the disentanglement of uncertainty types, such as active learning in predictive maintenance, radial BNNs and MC dropout appear to be superior options compared to heteroscedastic neural networks, which, despite their strong results, do not naturally separate uncertainty sources [48]. Moreover, the trade-offs between computational cost and uncertainty quality are evident across various studies. While MCMC provides the highest quality uncertainty estimates, its computational burden often renders it infeasible for large-scale models, leading practitioners to favor variational inference or ensemble methods [14]. Ultimately, the literature

underscores that selecting a UQ method requires a nuanced understanding of the specific requirements of the application, including the need for calibration, the availability of data, and the criticality of distinguishing between different sources of uncertainty [3].

Emerging Trends, Challenges, and Future Directions

The trajectory of Bayesian Deep Learning (BDL) is undergoing a paradigm shift, moving from a primary focus on accurate uncertainty quantification (UQ